The molecular essence of Indian cuisine



Ganesh Bagler
Center for Computational Biology, IIIT-Delhi



Cooking is alchemy



Cooking: Central to evolution of human brain

Cooking is the essence of being human



‘Catching Fire—How cooking made us human’ by Richard Wrangham

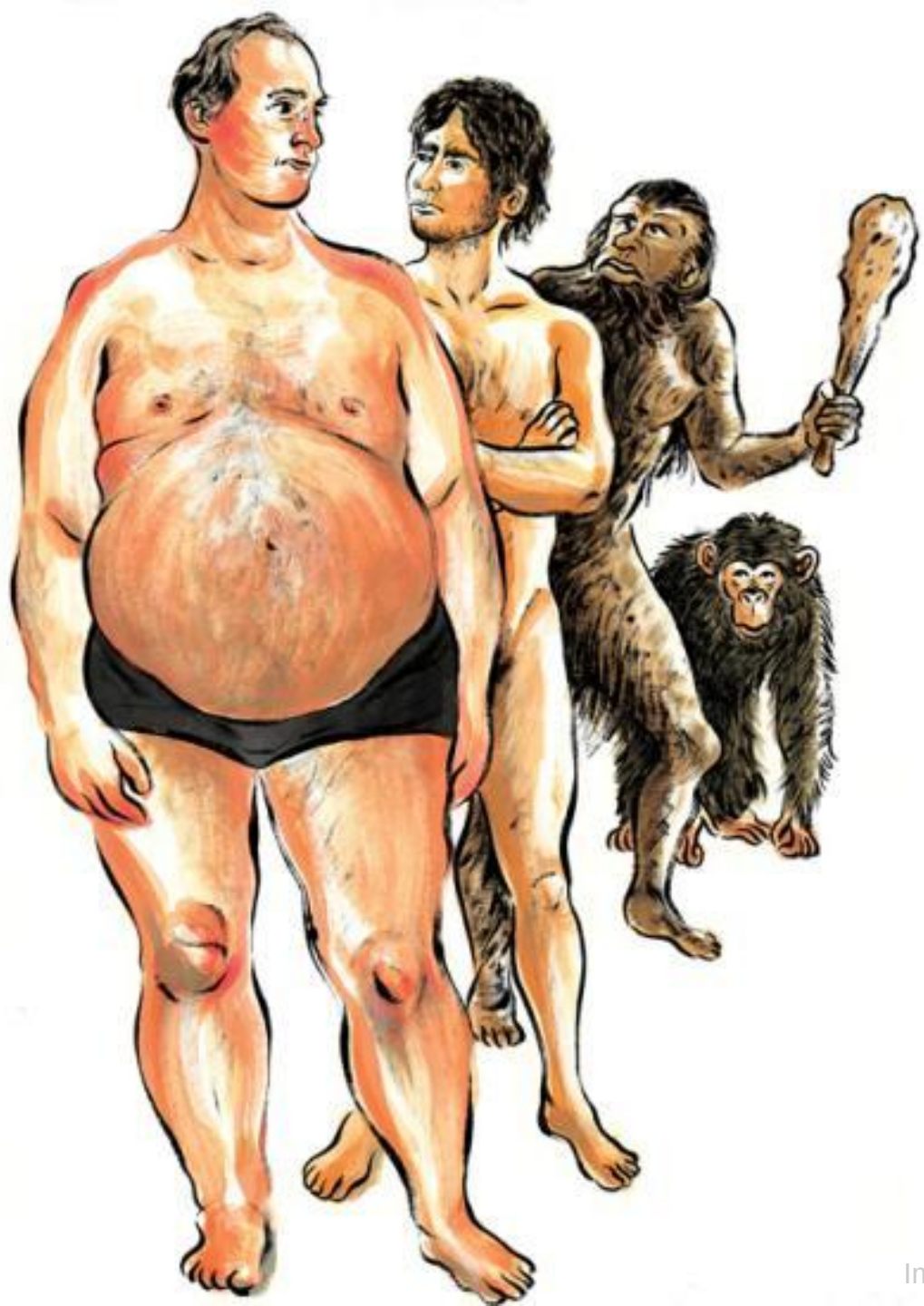


Image Credits: Chloe Cushman



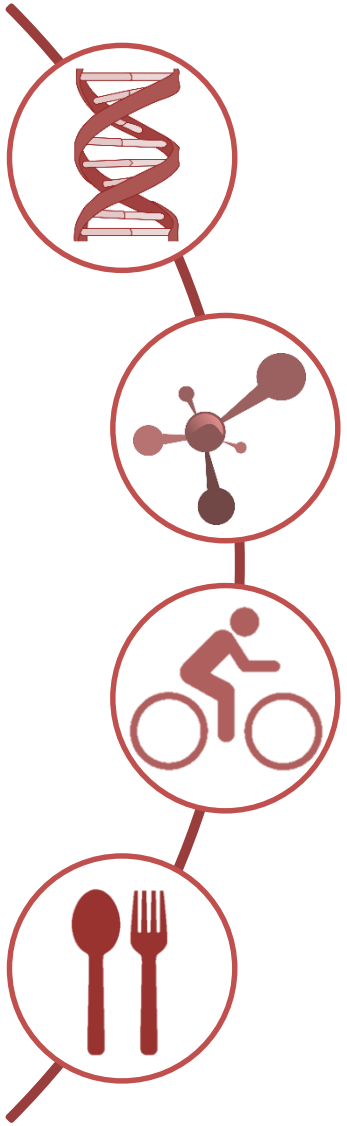
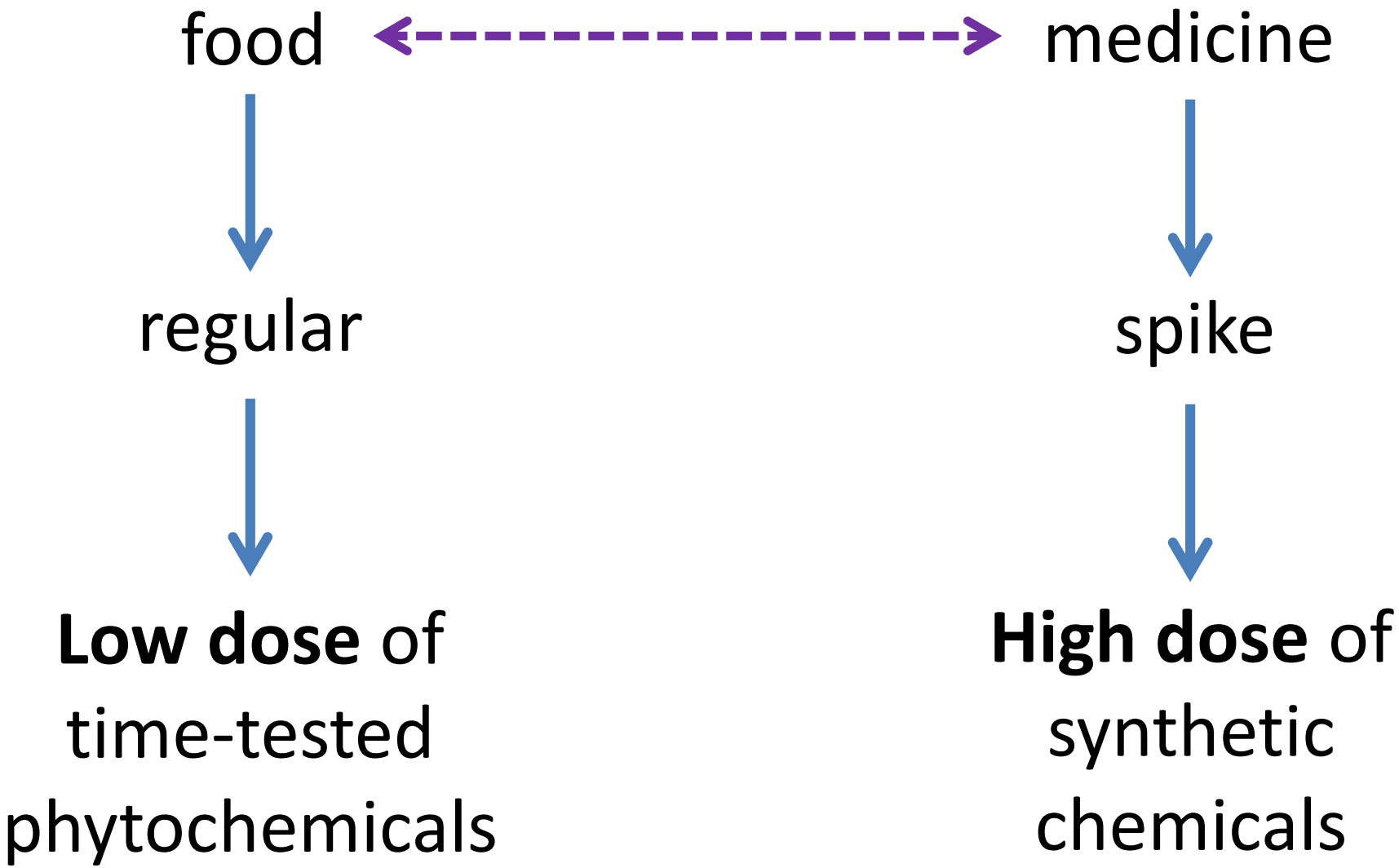


Image Credits: Chloe Cushman

diabetes

cancer


food
medicine
regular
spike
Low dose of time-tested phytochemicals
High dose of synthetic chemicals

Why do we eat what we eat?

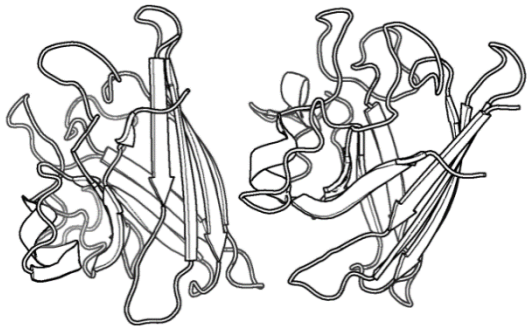
Why do we combine ingredients in our recipes the way we do?

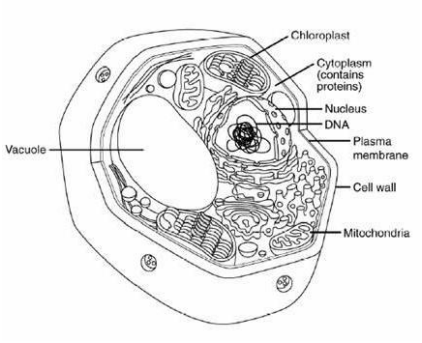


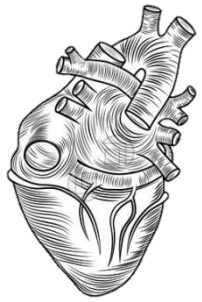
Recipe as a Complex System

A system comprising of large number of sophisticated functional elements,
intricately connected with each other to perform specific tasks,
which otherwise can not be executed by subsets of the system.

SUM OF PARTS
– IS NOT THE SAME AS –
THE WHOLE





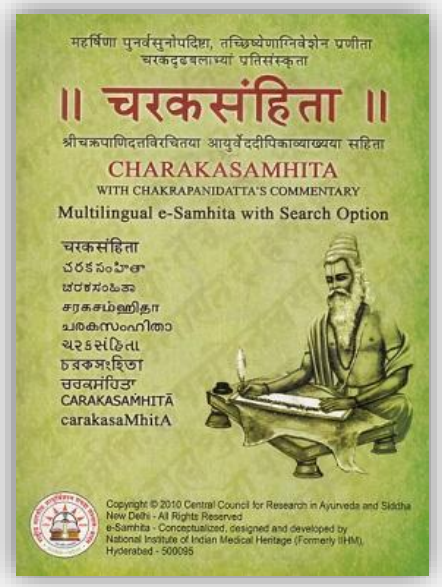


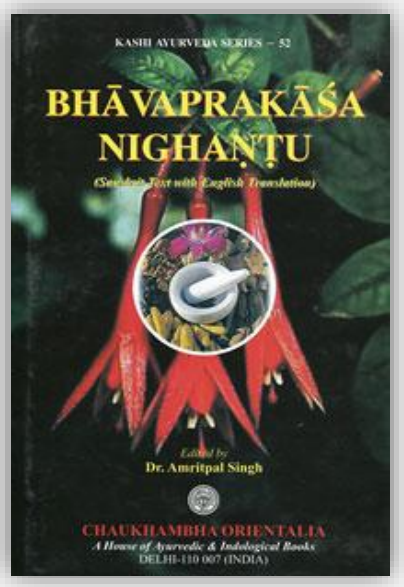


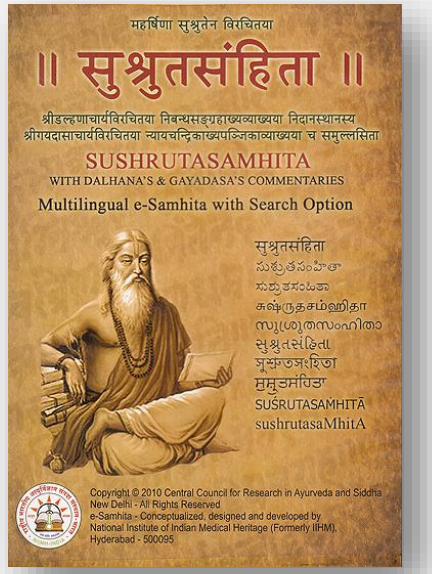
अन्नं वै प्राणाः

- India has a culinary history of **health-centric dietary practices** aimed at disease prevention and promotion of health.
- **Food as a medicine**: Traditionally, food has been treated as a medicinal agent in the Indian subcontinent.

Charaka Samhitā (चरक संहिता)
Susruta Samhitā (सुश्रुत संहिता)
Bhāvaprakāśa Nighaṇṭu (भावप्रकाश निघण्टु)







:: Data ::

well tested 

&

well tasted 

DATA

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2543 recipes, 194 ingredients, From 8 sub-cuisines—

Bengali, Gujarati, Jain, Maharashtrian, Mughlai, Punjabi, Rajasthani, South Indian



Recipe Contest of the Month

Recipes Using Sesame Seeds (til)

View contest archive....

New Release



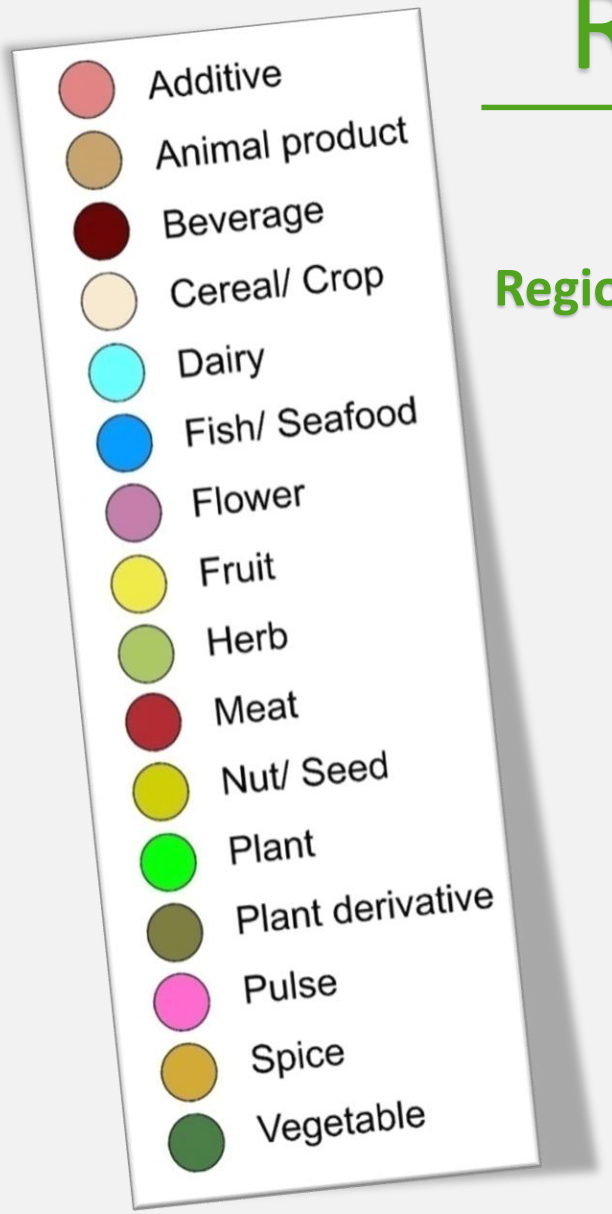


- Carrot and Spinach Soup
- Cucumber Cooler
- Rajma Cheese Paratha
- Aloo Paneer Chaat

Bright Vision! | Green Garlic! | Eat the Italian Way | Soft and Fluffy Pancakes

PAUSE

Recipes & Ingredients



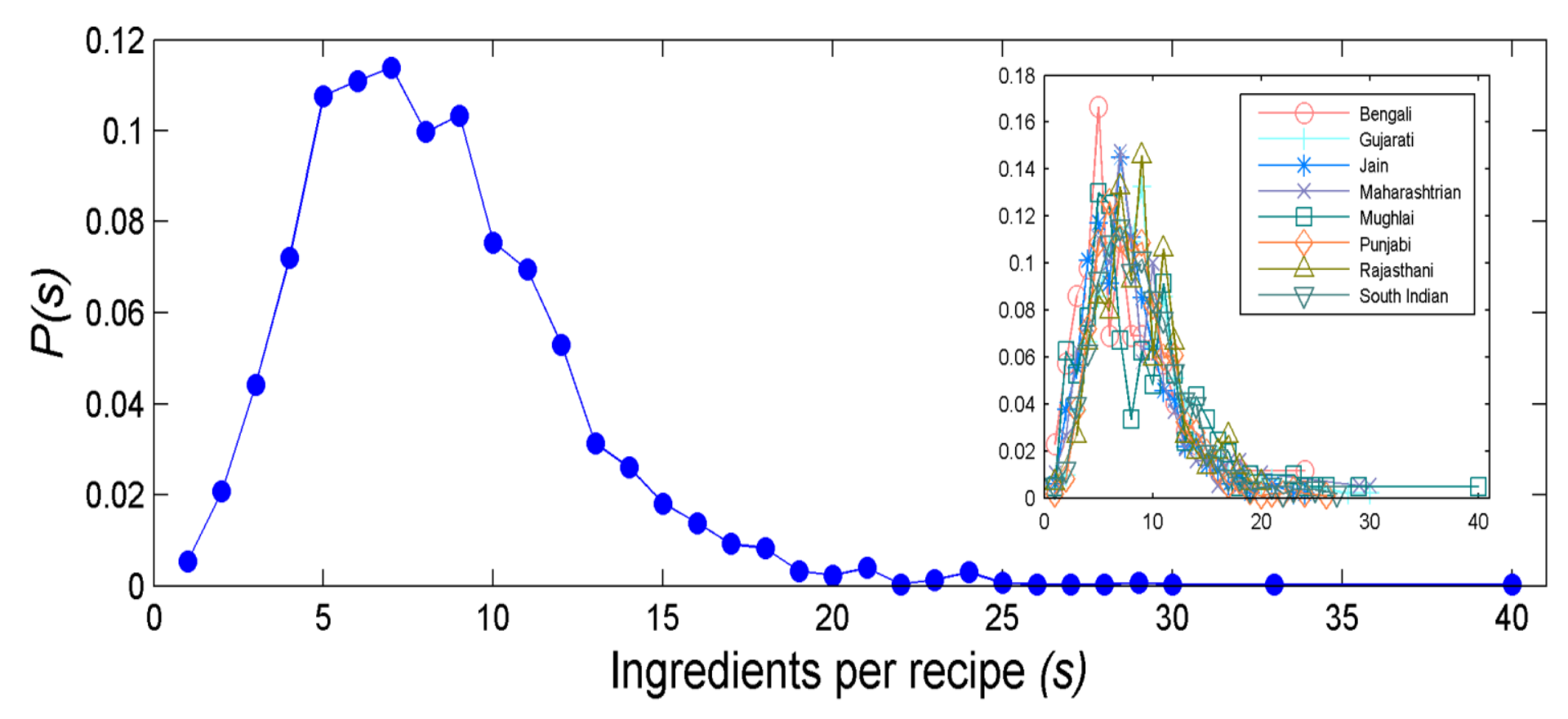
2543 Traditional Indian Recipes (TarlaDalal)

Regional cuisines: Bengali, Gujarati, Jain, Maharashtrian, Mughlai, Punjabi, Rajasthani, South Indian.



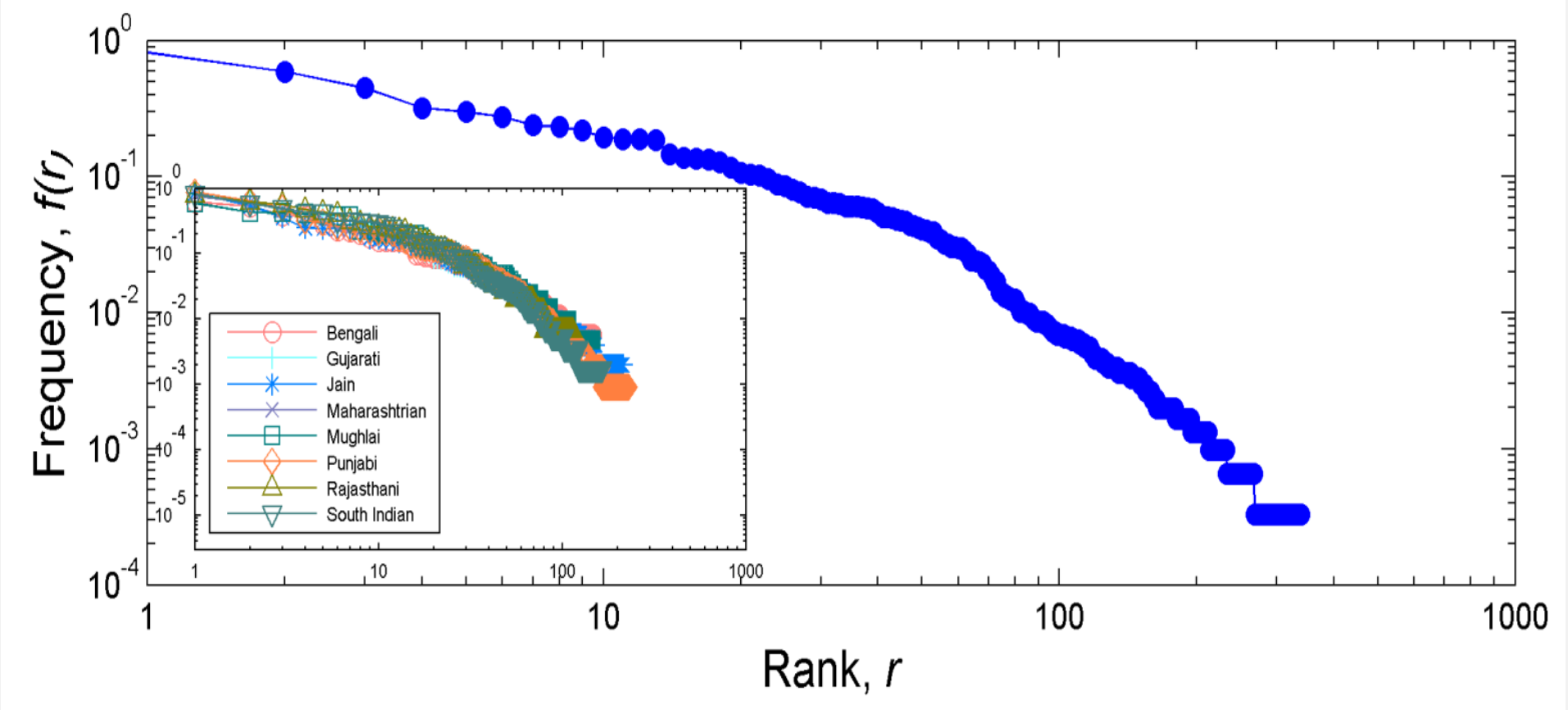
A Jain, NK Rakhi & G Bagler,* "Analysis of Food Pairing in Regional Cuisines of India", **PLoS ONE**, *10(10): e0139539*(2015).

Indian cuisine: Recipe size statistics



A Jain, NK Rakhi, G Bagler, “Spices form the basis of food pairing in Indian cuisine”, arXiv:1502.03815 (2015).

Indian cuisine: Frequency of use of ingredients



- Scale free distribution.
- Invariant across sub-cuisines.

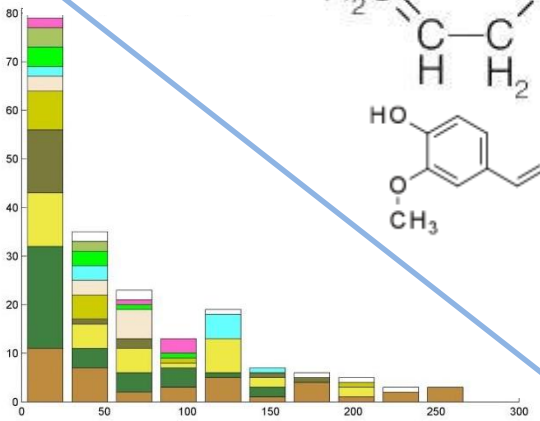
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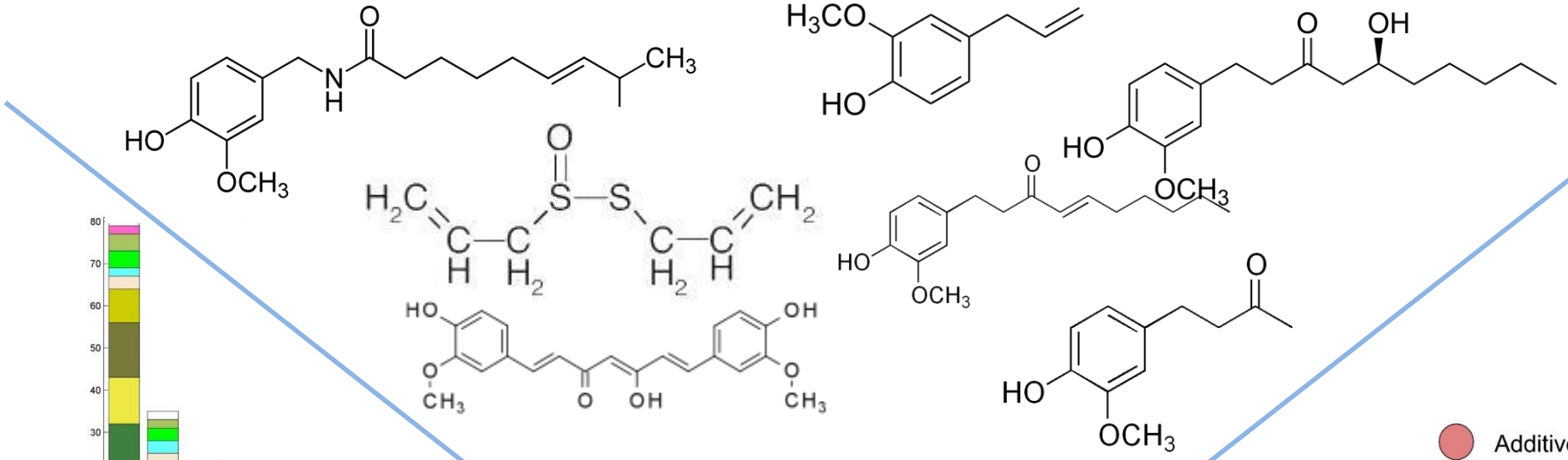
olfactory

fla • vor = aroma + taste

gustatory

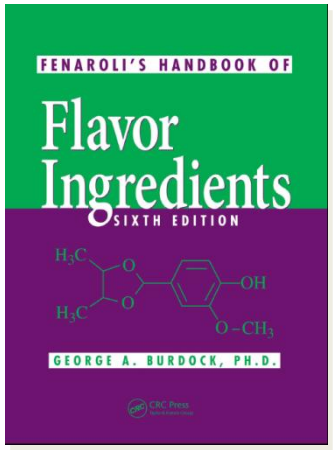






1170 flavor molecules

PubChem



194 ingredients

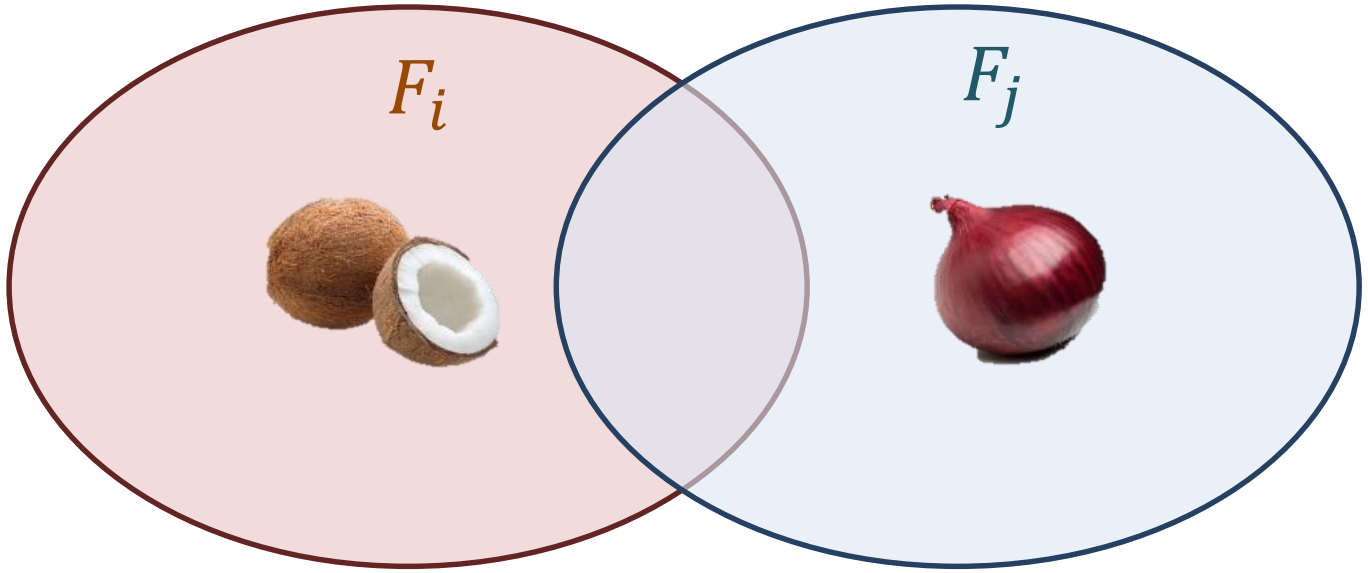
Ahn et. al, *Sci.Rep.,* 1, 196 (2011).

- Additive
- Animal product
- Beverage
- Cereal/ Crop
- Dairy
- Fish/ Seafood
- Flower
- Fruit
- Herb
- Meat
- Nut/ Seed
- Plant
- Plant derivative
- Pulse
- Spice
- Vegetable



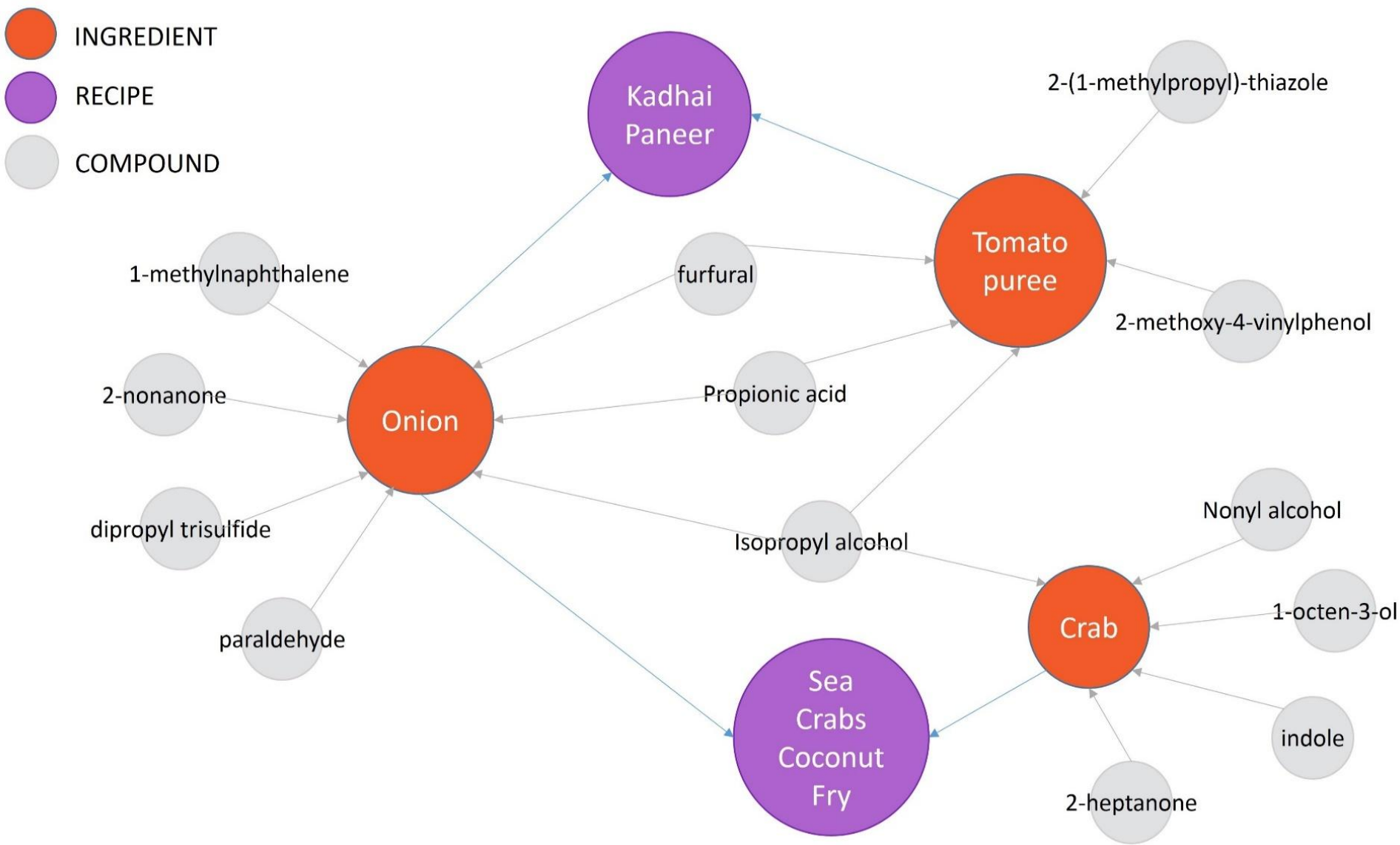
Food Pairing Hypothesis

Ingredients sharing flavor compounds are more likely to taste well together than ingredients that do not.

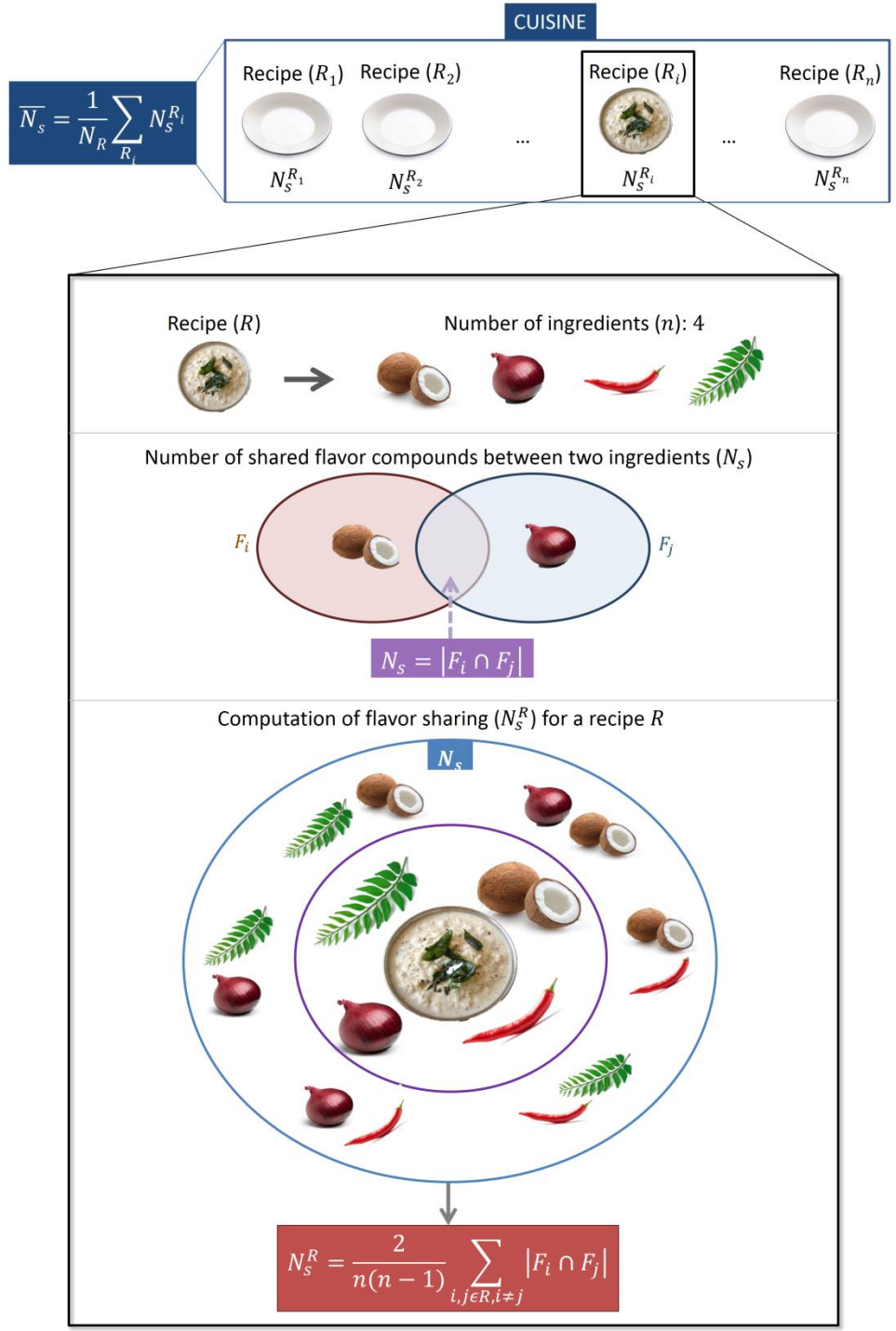


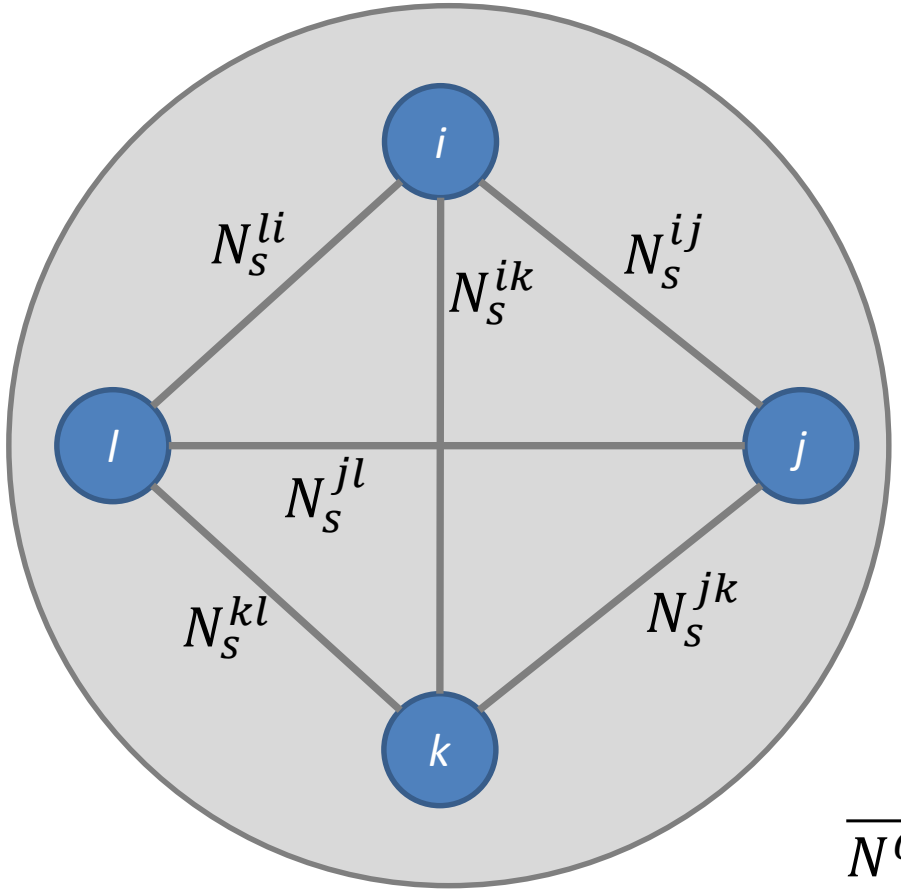
$$N_s = |F_i \cap F_j|$$

A Jain, NK Rakhi, G Bagler, “Spices form the basis of food pairing in Indian cuisine”, arXiv:1502.03815 (2015).


INGREDIENT
RECIPE
COMPOUND
Kadhai Paneer
2-(1-methylpropyl)-thiazole
Tomato puree
1-methylnaphthalene
furfural
2-methoxy-4-vinylphenol
2-nonanone
Onion
Propionic acid
dipropyl trisulfide
Isopropyl alcohol
Nonyl alcohol
Crab
1-octen-3-ol
paraldehyde
Sea Crabs Coconut Fry
indole
2-heptanone

Food Pairing





R

N_s^{ij} Number of shared flavor molecules between ingredients *i* and *j*

$\overline{N_s^R}$ Average number of shared flavor molecules across all pairs of ingredients in the recipe R

$\overline{N^{Cuisine}}$ Average number of shared flavor molecules across all recipes in a cuisine

Ahn *et. al, Sci.Rep.* (2011)

Uniform Food Pairing

Western Cuisines

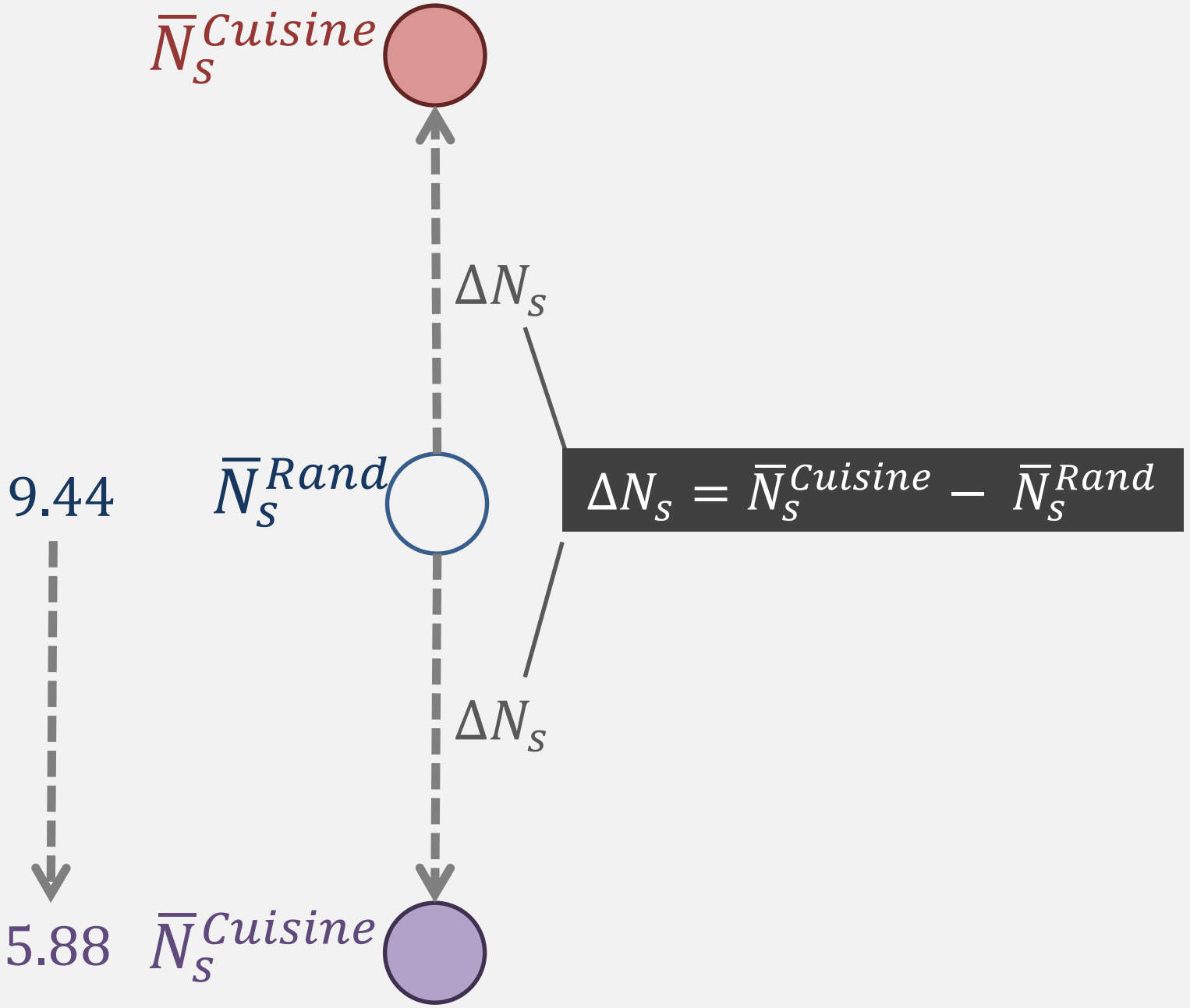


Random Cuisine

Indian Cuisine

Contrasting Food Pairing

Jain *et. al*, *PLoS ONE* (2015)


$\overline{N}_s^{Cuisine}$
ΔN_s
$\overline{N}_s^{Rand}$
9.44
$\Delta N_s = \overline{N}_s^{Cuisine} - \overline{N}_s^{Rand}$
ΔN_s
5.88
$\overline{N}_s^{Cuisine}$

Indian cuisine is characterized with negative food pairing.

more the extent of flavor sharing between any two ingredients, lesser their co-occurrence

Western cuisines have a *uniform* blend of ingredients.

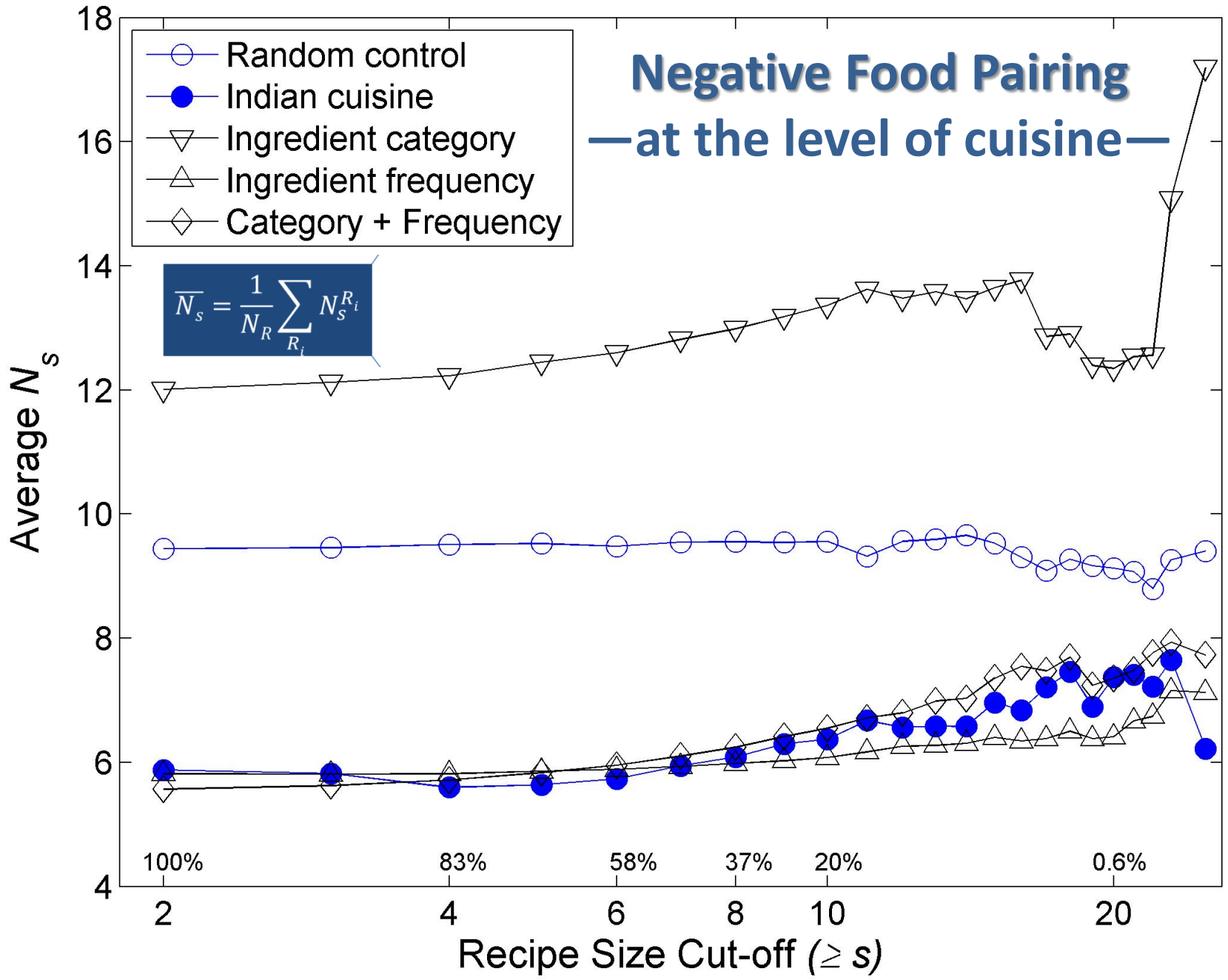
Indian cuisine has a *contrasting* blend of ingredients.

How to create a random cuisine? (Strategy – 1)

(1) Corresponding to every recipe, create its size-controlled randomized version by randomly sampling ingredients from the 'Ingredients Basket' (without replacement).
(2) Having thus created 'a randomized cuisine', create a large number of such cuisines (say, 100) for statistical analysis.



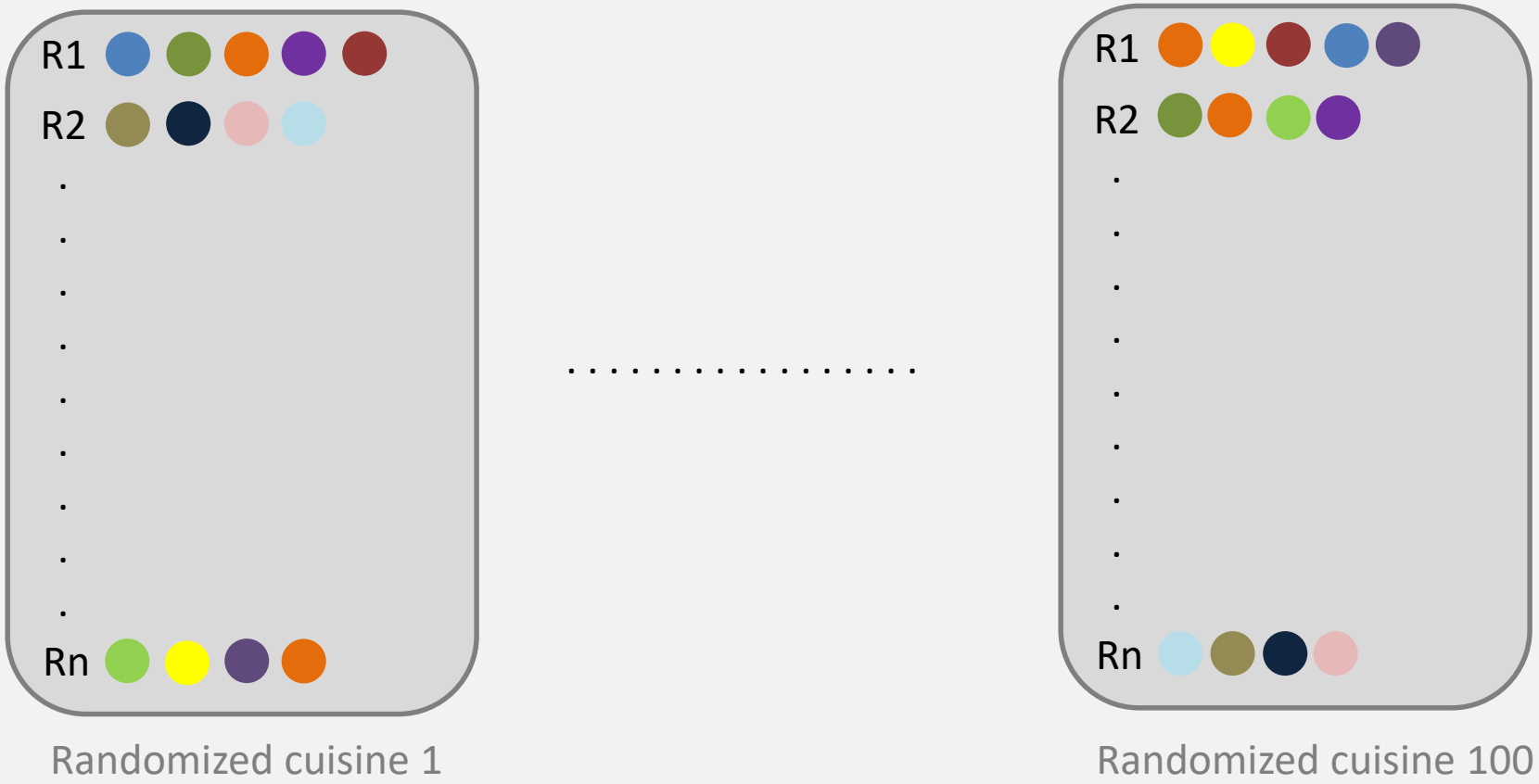
Randomize cuisine 1 . . . . . Randomize cuisine 100


Negative Food Pairing
—at the level of cuisine—
Random control
Indian cuisine
Ingredient category
Ingredient frequency
Category + Frequency
$\overline{N_s} = \frac{1}{N_R}\sum_{R_i} N_s^{R_i}$
Average N_s
Recipe Size Cut-off ($\geq s$)
100%
83%
58%
37%
20%
0.6%
18
16
14
12
10
8
6
4
2
4
6
8
10
20

How to create a random cuisine by PRESERVING CATEGORY COMPOSITION?

(1) Corresponding to every recipe, create its size-controlled randomized version by sampling ingredients from the 'Ingredients Basket' (without replacement) BY PRESERVING/MAINTAINING ITS CATEGORY.

(2) Having thus created 'a randomized cuisine', create a large number of such cuisines (say, 100) for statistical analysis.

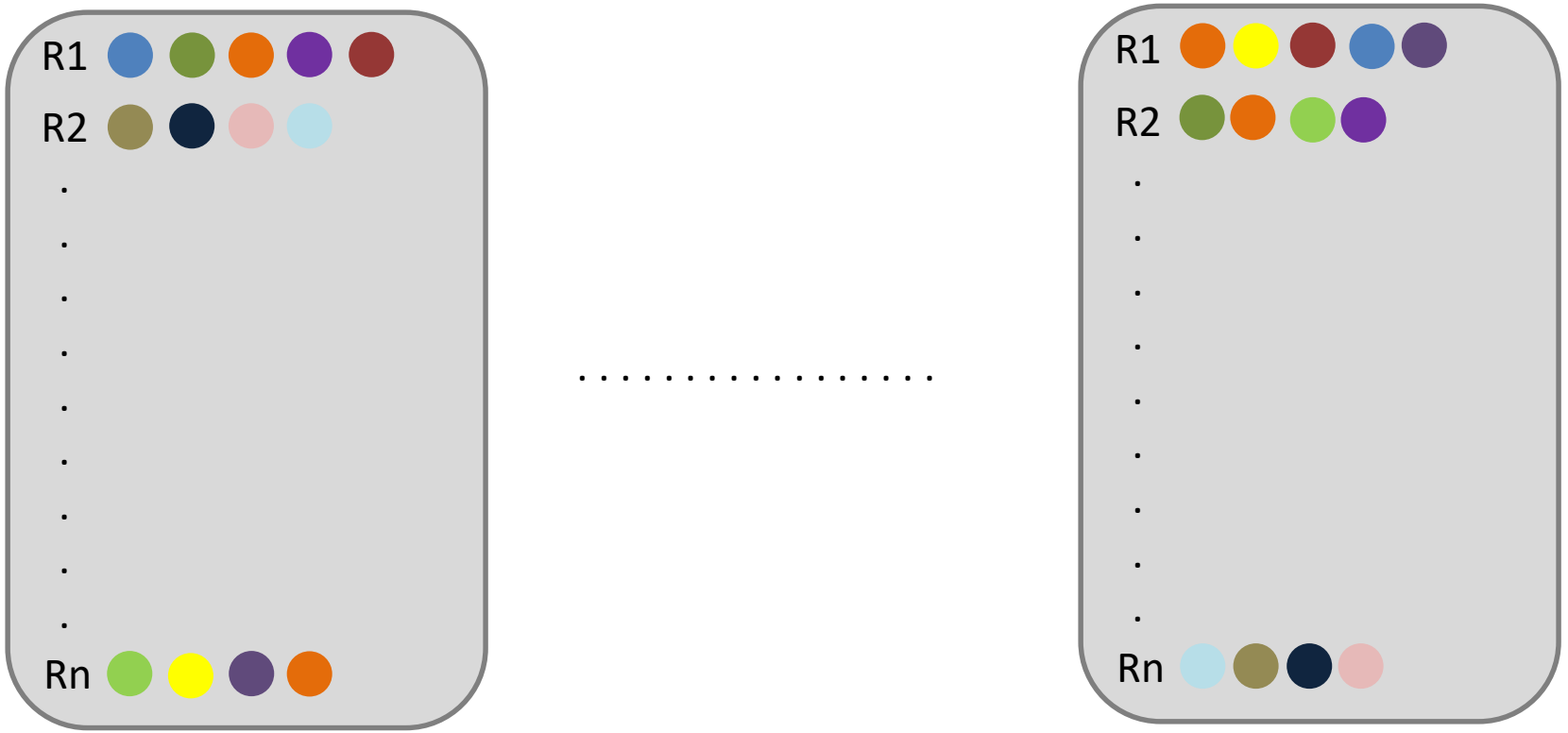


Randomized cuisine 1 . . . . . Randomized cuisine 100

How to create a random cuisine by PRESERVING INGREDIENT FREQUENCY?

(1) Corresponding to every recipe, create its size-controlled randomized version by sampling ingredients from the 'Ingredients Basket' (without replacement) SUCH THAT INGREDIENTS ARE PICKED PROPORTIONAL TO THEIR FREQUENCY.

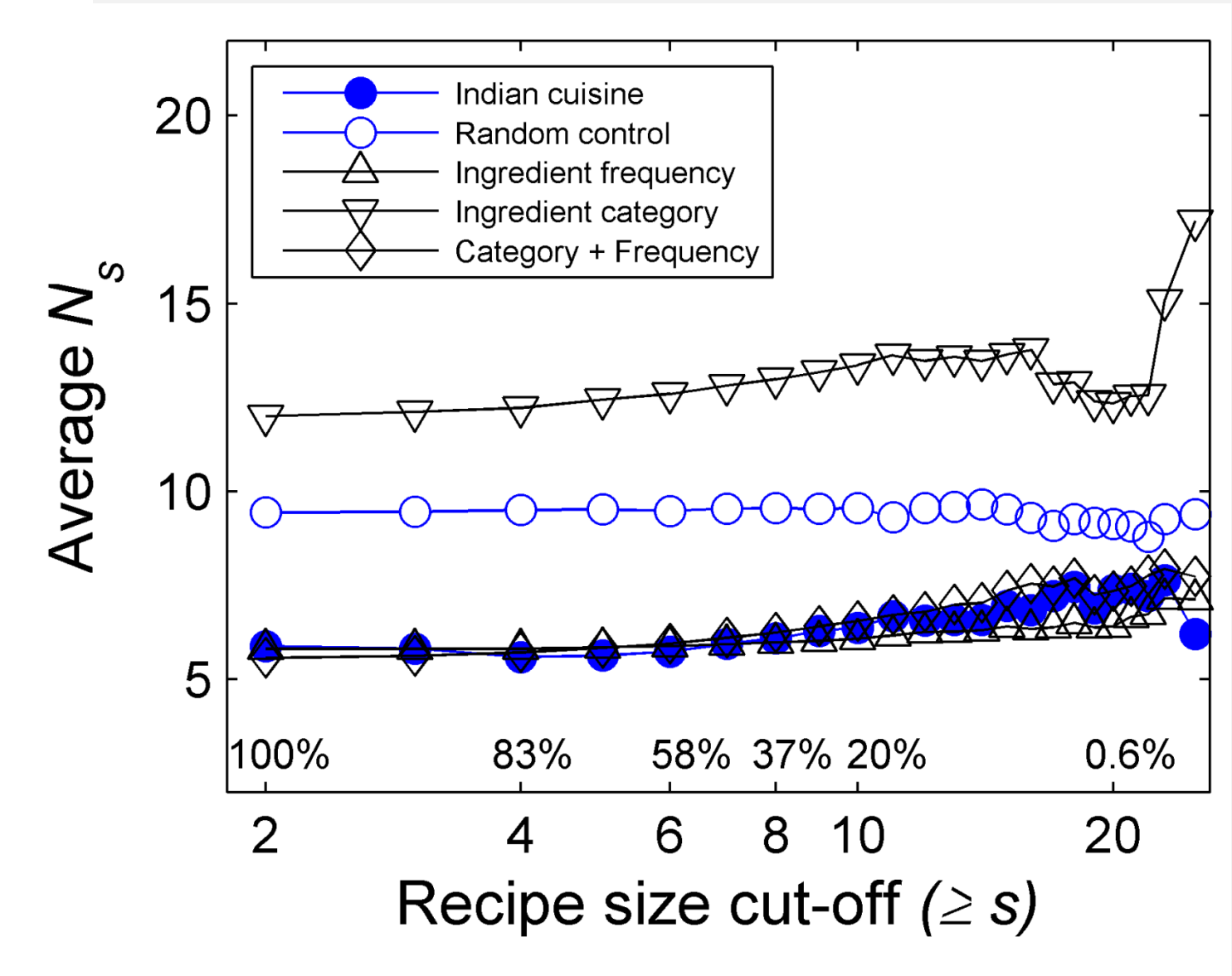
(2) Having thus created 'a randomized cuisine', create a large number of such cuisines (say, 100) for statistical analysis.



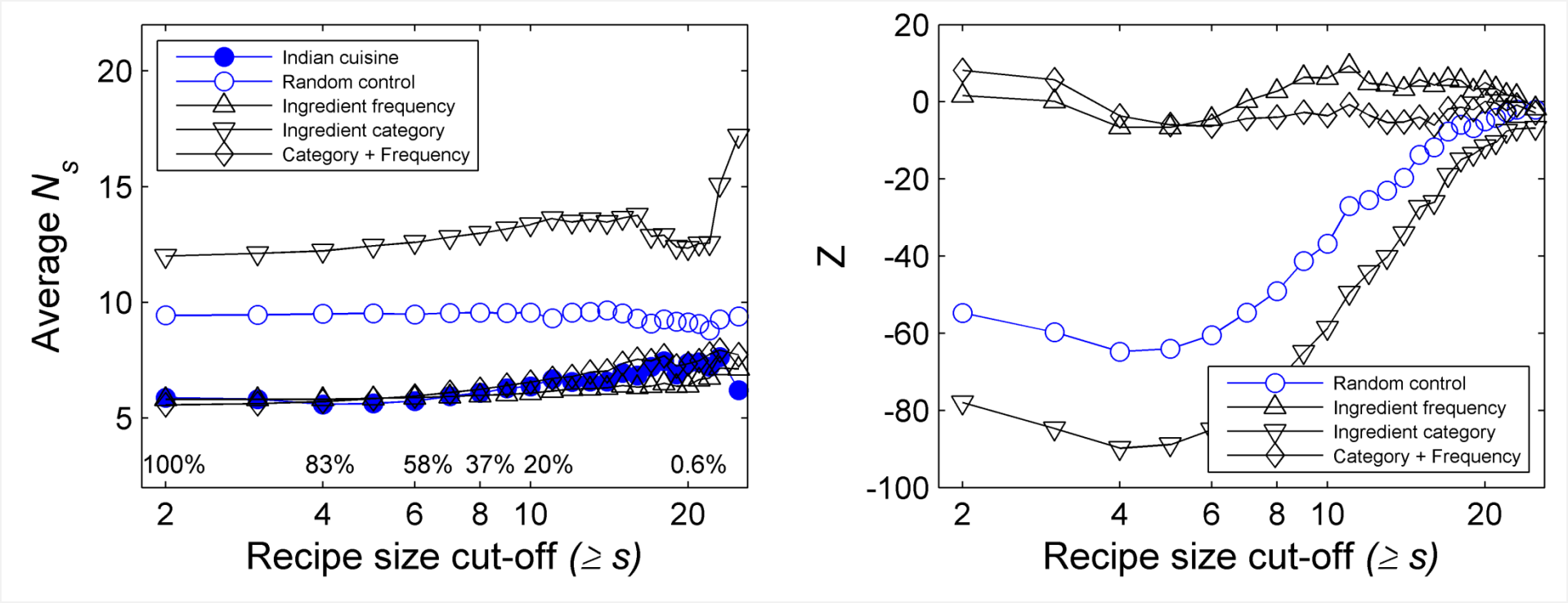
Randomized cuisine 1

Randomized cuisine 100

Negative Food Pairing
—at the level of cuisine—



Negative Food Pairing
—at the level of cuisine—

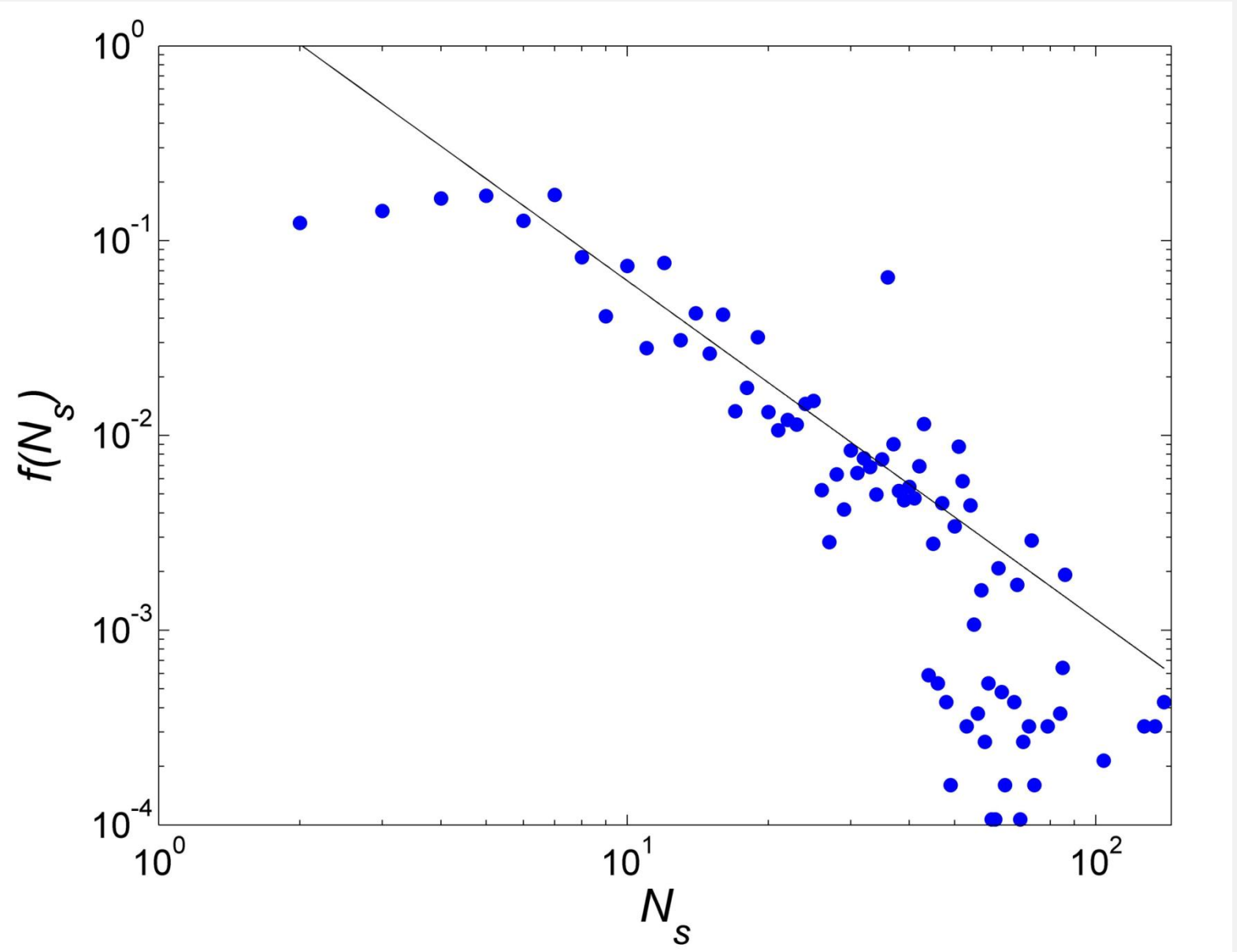


Number of recipes in random controls: 20,000

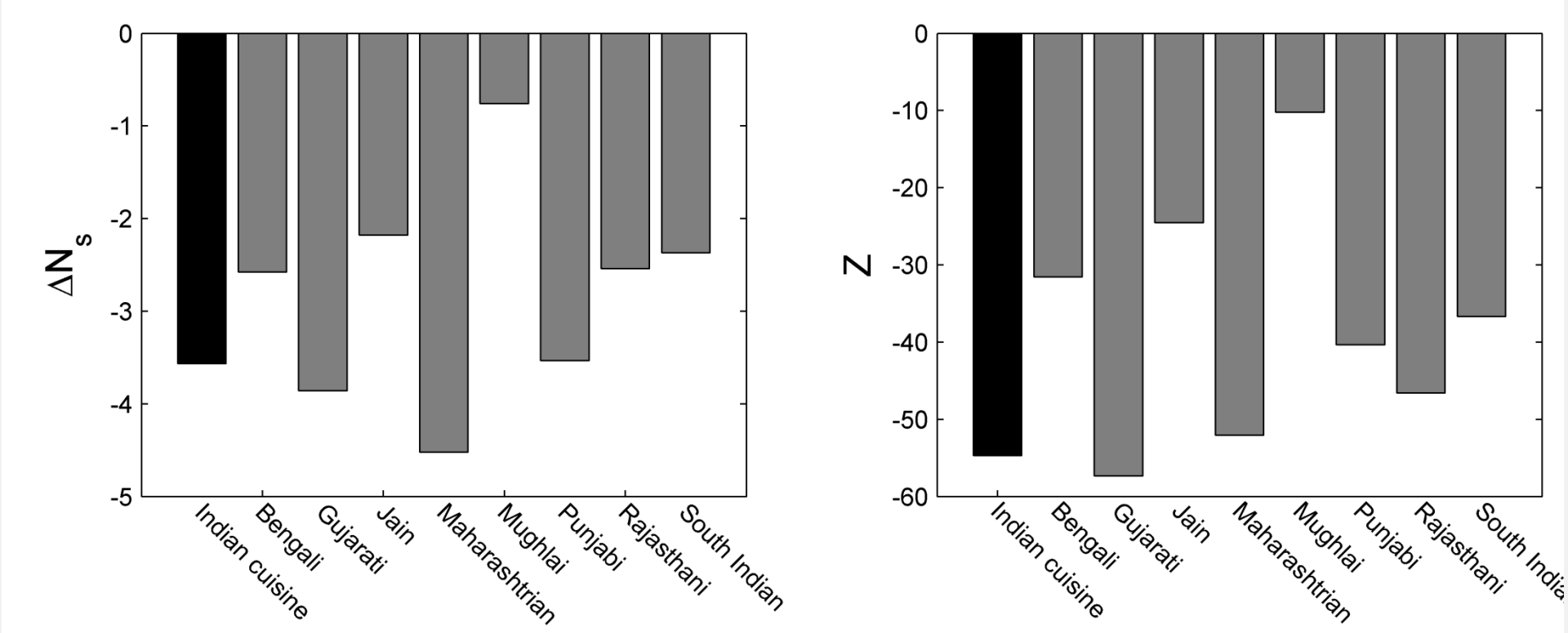
$$Z = \sqrt{N_{Rand}} \frac{(\overline{N}_s^{IC} - \overline{N}_s^{Rand})}{\sigma_{Rand}}$$

A Jain, NK Rakhi, G Bagler, "Spices form the basis of food pairing in Indian cuisine", arXiv:1502.03815 (2015).

Negative Food Pairing
—at the level of ingredient pairs—

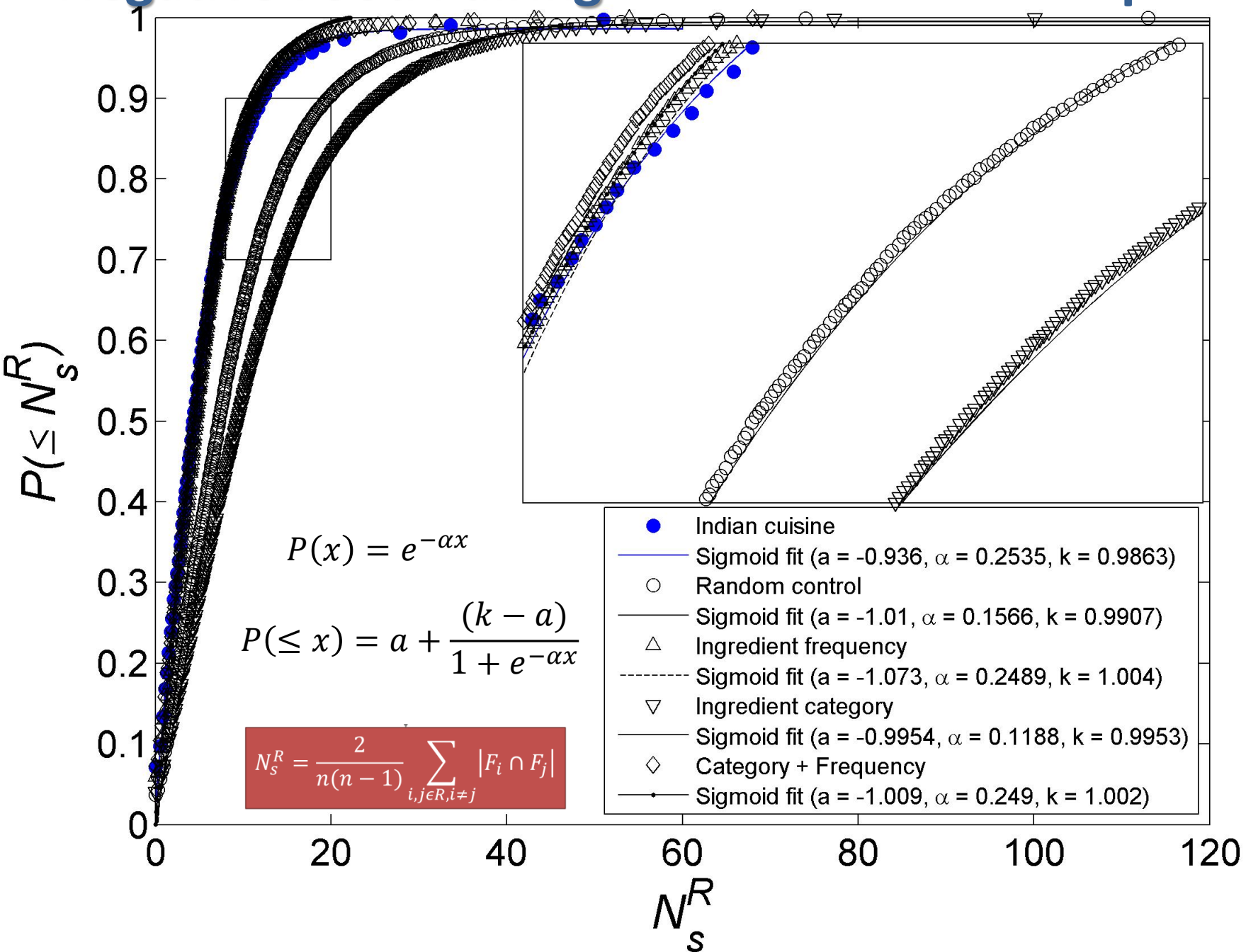


Negative Food Pairing
—at the level of sub-cuisines—



A Jain, NK Rakhi, G Bagler, “Spices form the basis of food pairing in Indian cuisine”, arXiv:1502.03815 (2015).

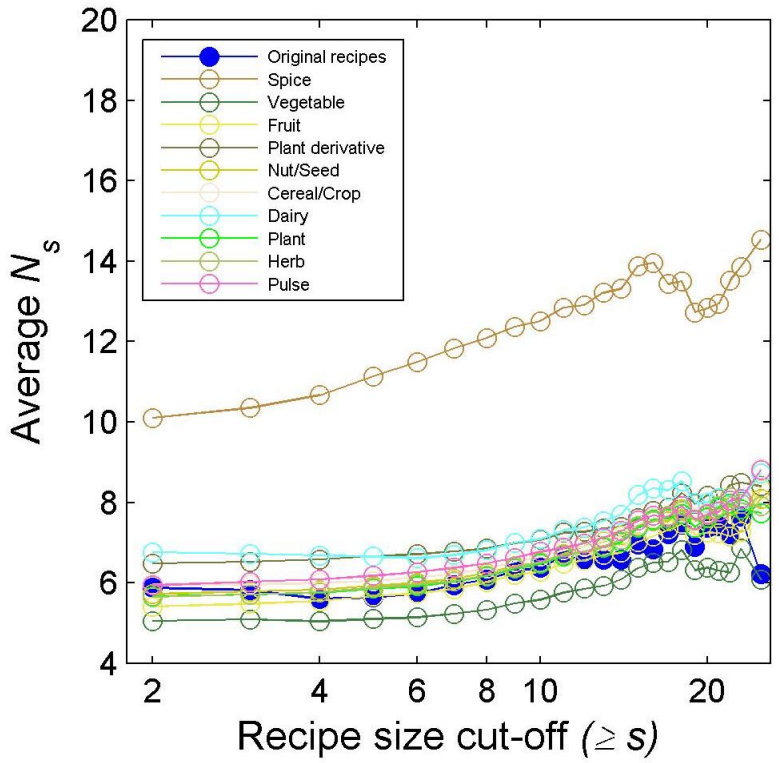
Negative Food Pairing —at the level of recipes—

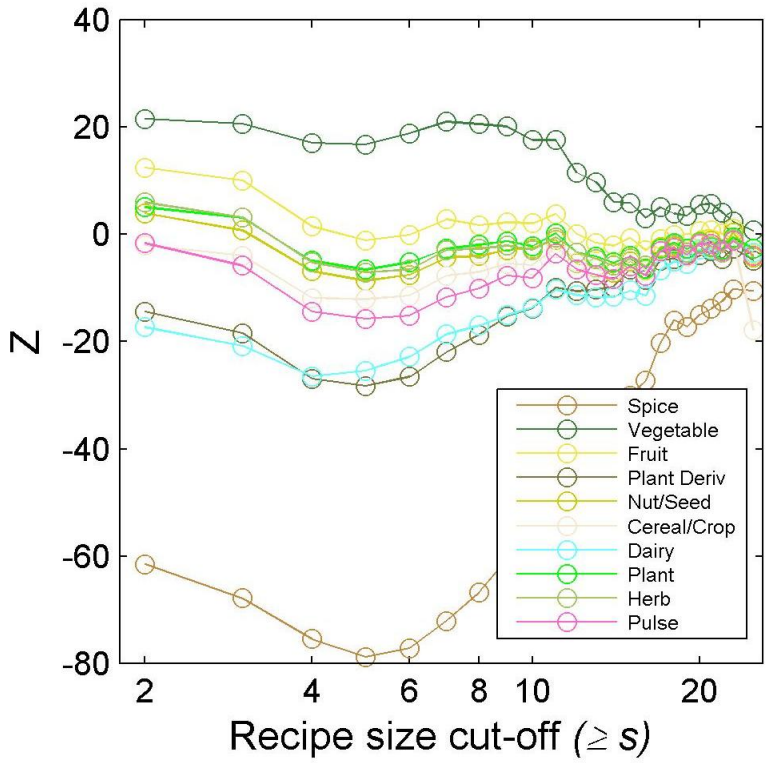


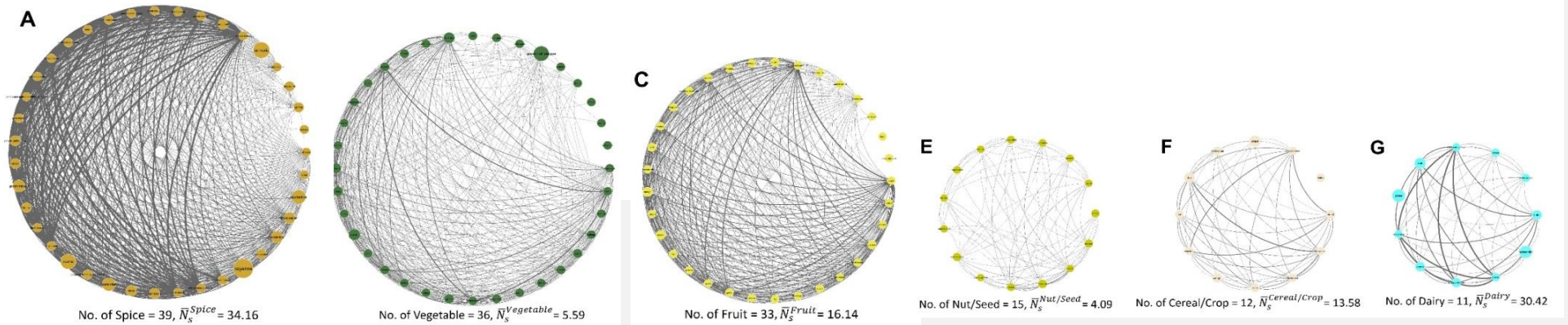
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more the extent of flavor sharing between any two ingredients, lesser their co-occurrence

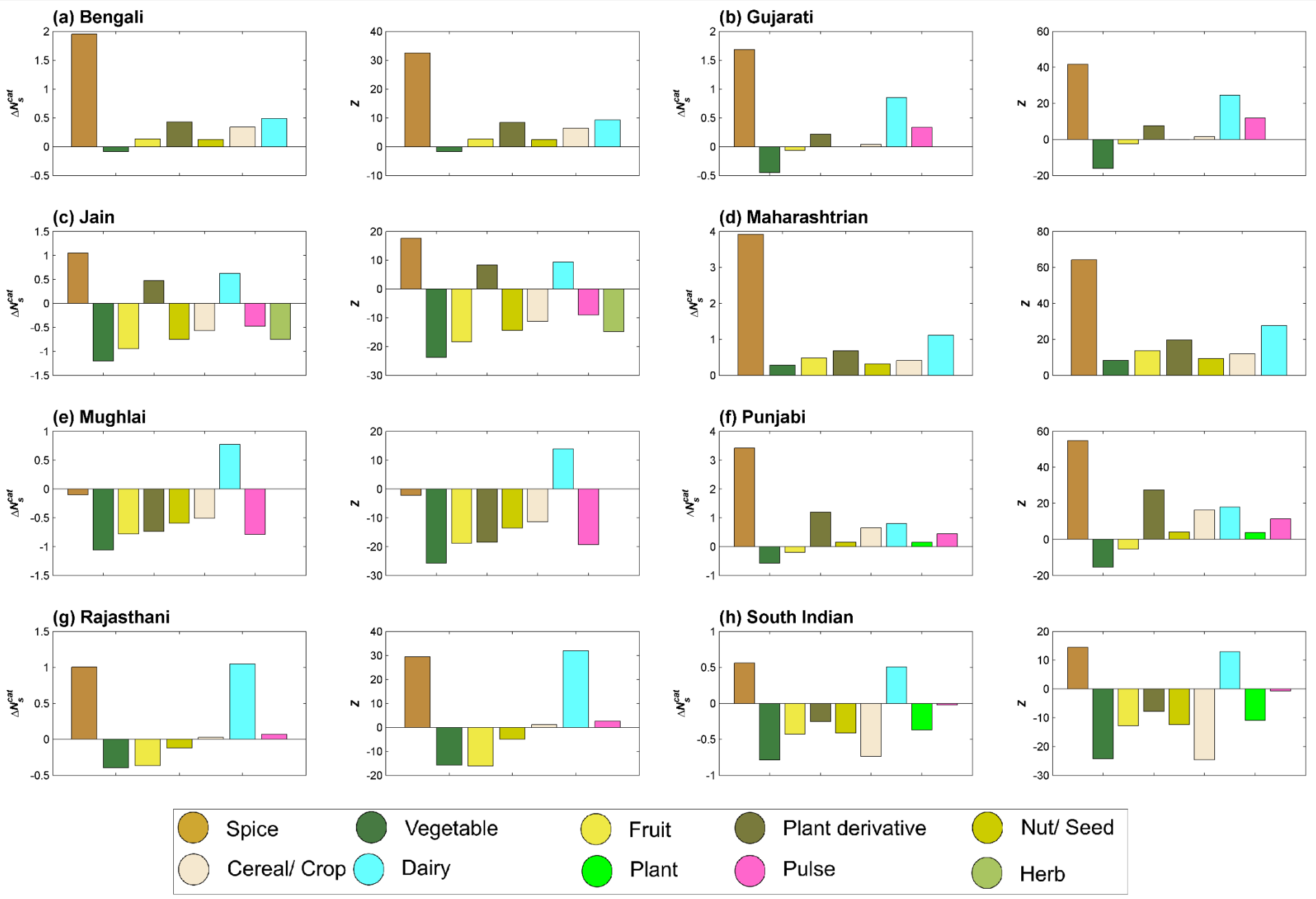
Spices are key to the food pairing in Indian cuisine



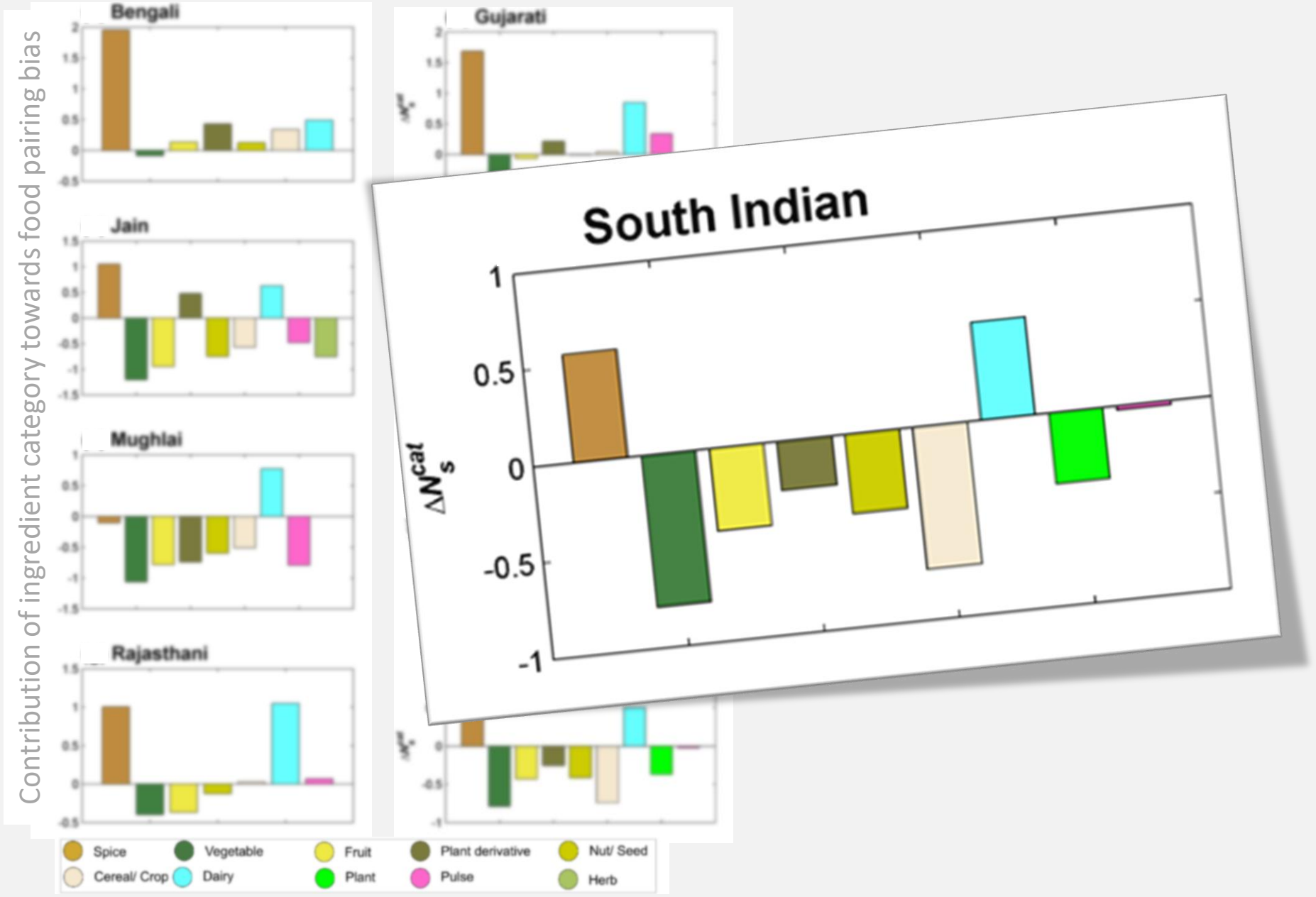




Culinary Fingerprinting of Regional Cuisines of India

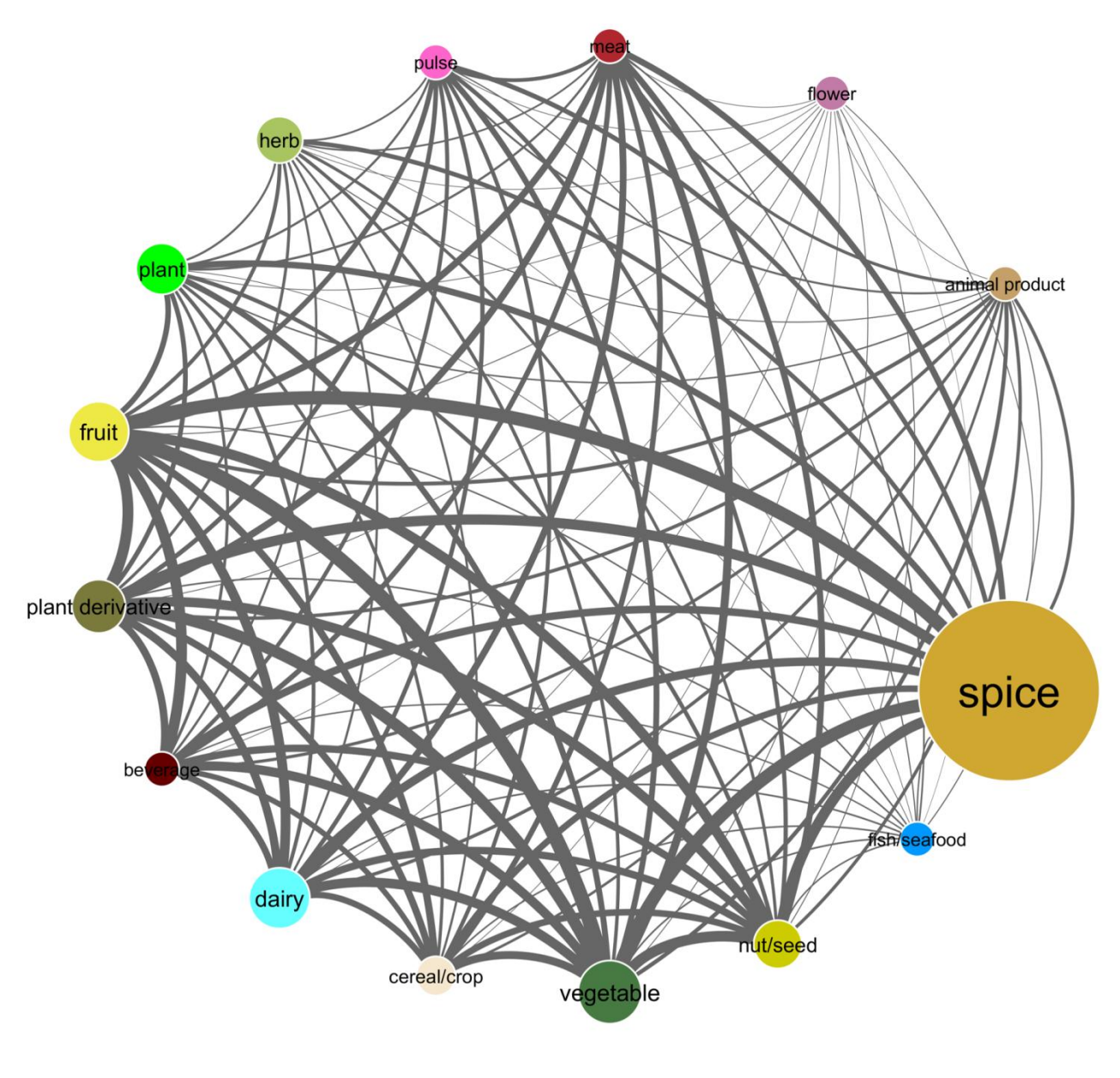


Culinary Fingerprints of Regional Cuisines of India



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Spices are key to the food pairing in Indian cuisine



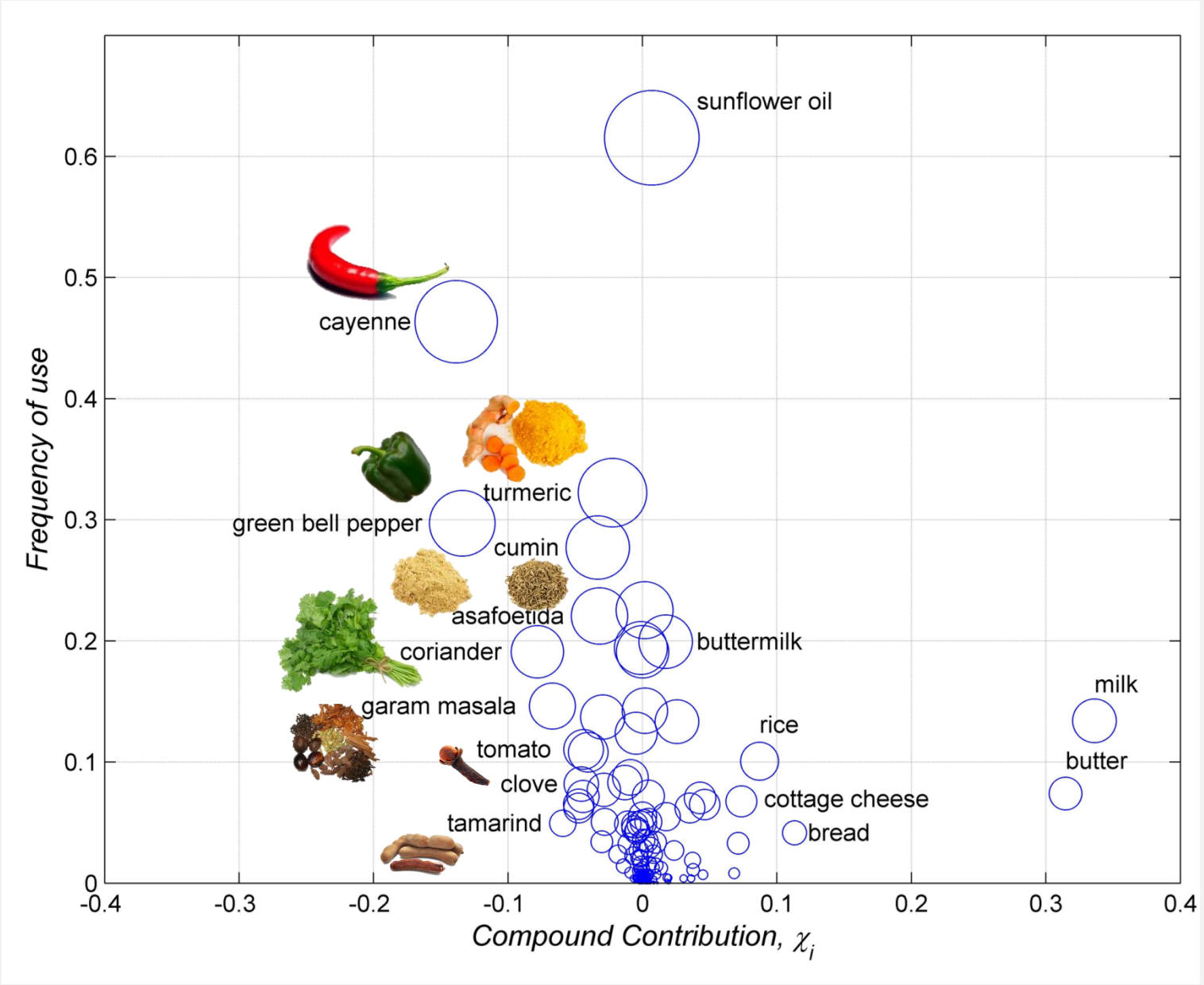
Role of ingredients in biasing the food pairing

$$+\infty \geq \chi_i \geq -\infty$$

$$\chi_i = \left(\frac{1}{N_R}\sum_{R \ni i}\frac{2}{n(n-1)}\sum_{j \neq i (j.i \in R))}\left|F_i \cap F_j\right|\right) - \left(\frac{2f_i}{N_R\langle n\rangle}\frac{\sum_{j\in c} f_j\left|F_i \cap F_j\right|}{\sum_{j\in c} f_j}\right)$$

A Jain, NK Rakhi, G Bagler, “Spices form the basis of food pairing in Indian cuisine”, arXiv:1502.03815 (2015).

Role of ingredients in biasing the food pairing











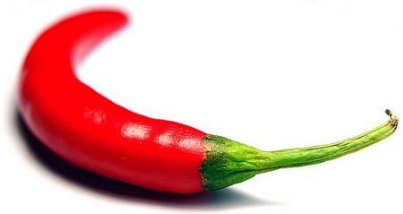












Spice

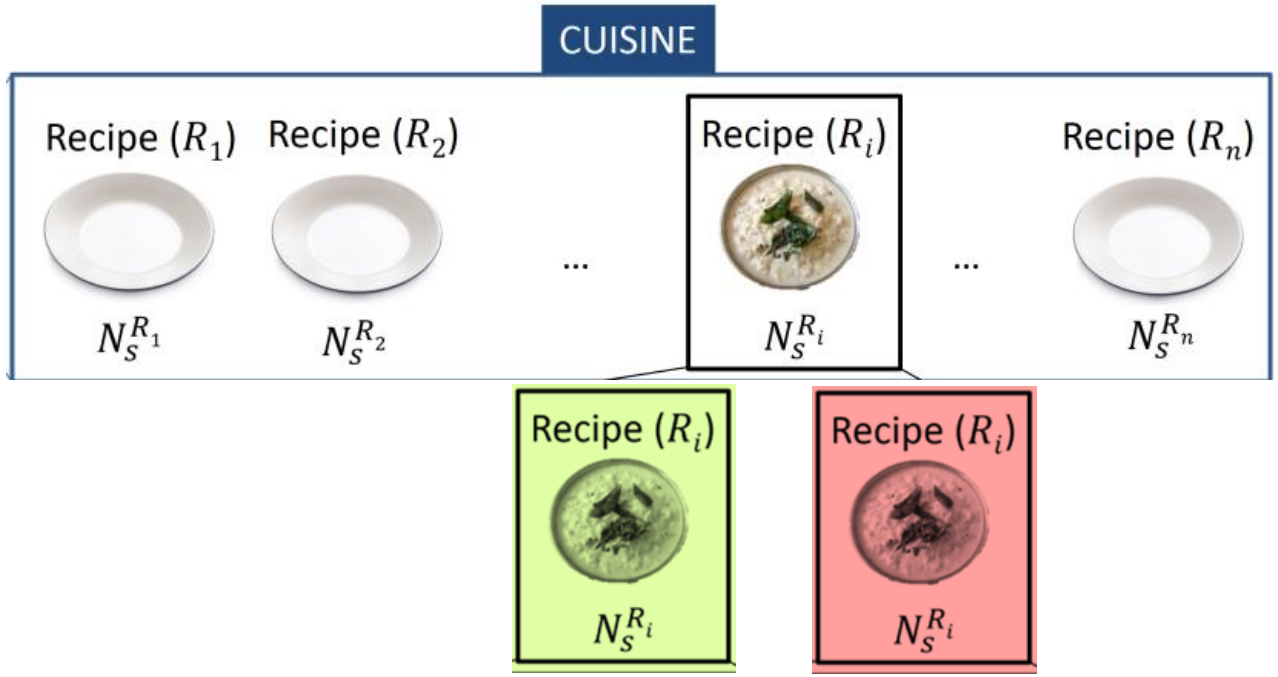
The Taste of India

भारत → बहारात





A Culinary Evolution Model

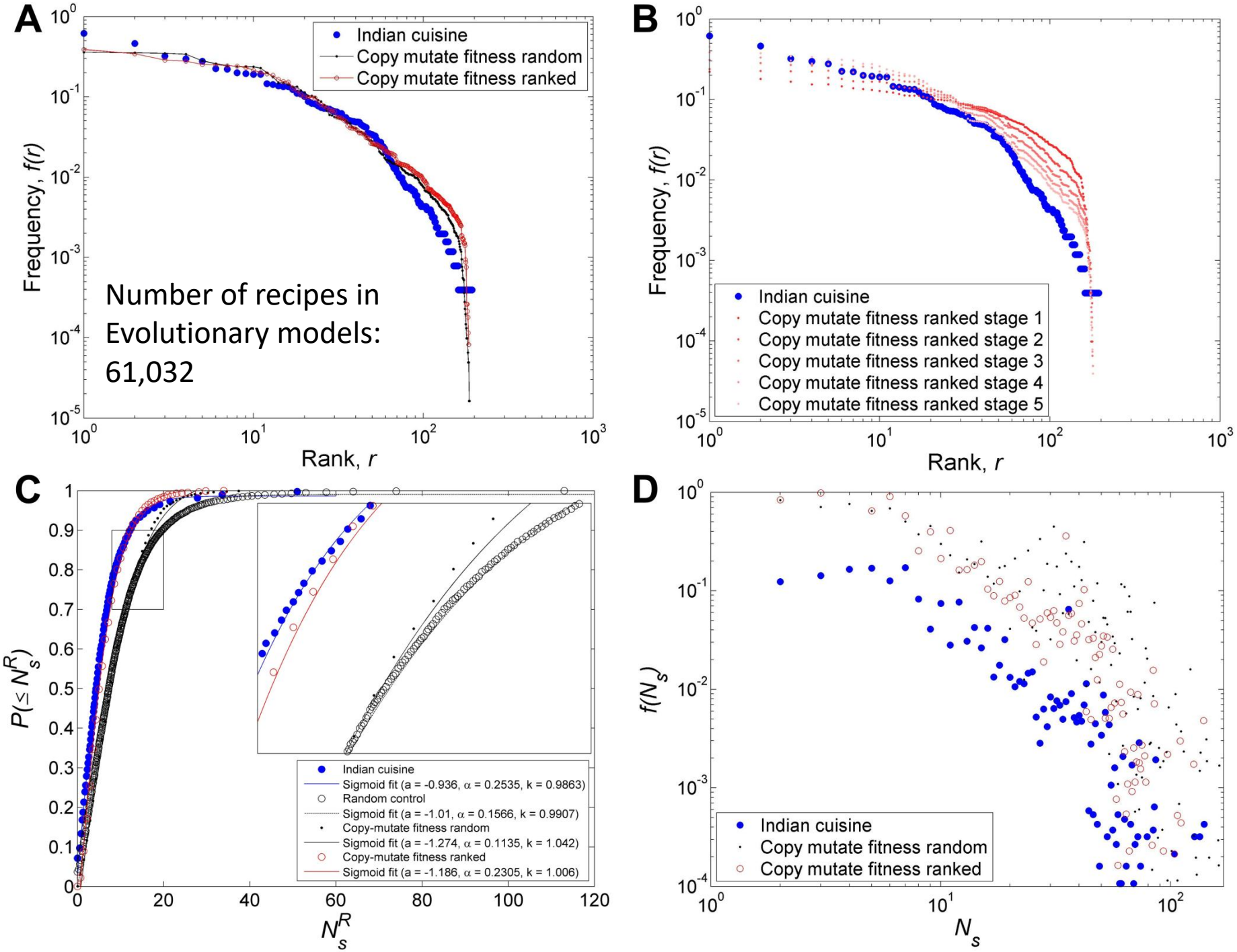


- Ascribe 'fitness values' to all the ingredients.

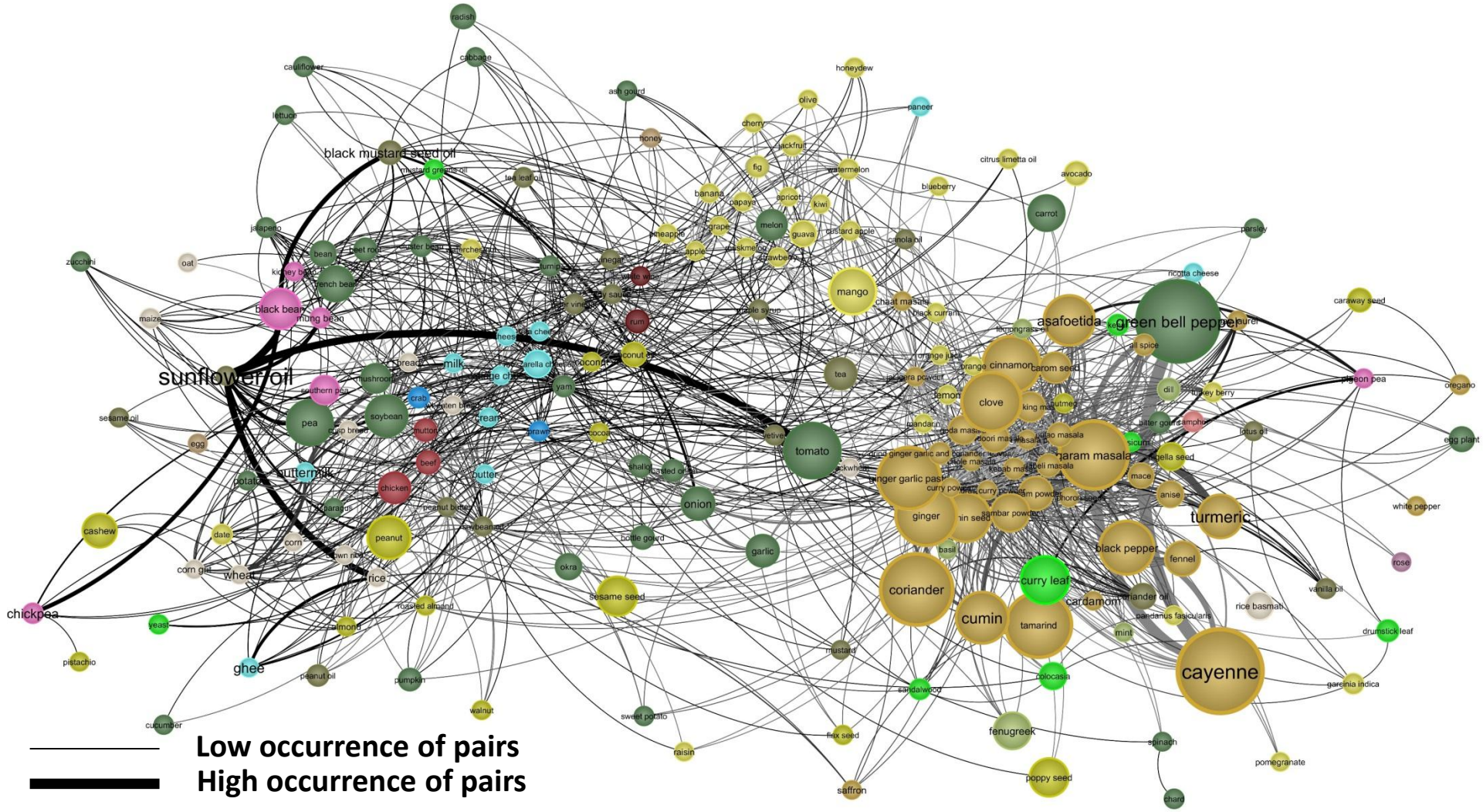
  (Model-1: Fitness Random; Model-2: Fitness Scaled)

- Start with a small 'ingredient pool'
- Create an initial set of recipes ($R_0 = 20$) of size K $(= 7)$ each.
- **Copy** a recipe randomly and **mutate** it L(=2) times.
- Maintain the size ratio of ingredient pool and recipe pool (M=0.0762; Indian cuisine).

Kinouchi et. al, *New J Phys* , 10 (7), 073020 (2008)


A
Indian cuisine
Copy mutate fitness random
Copy mutate fitness ranked
Number of recipes in Evolutionary models: 61,032
Frequency, f(r)
Rank, r
B
Indian cuisine
Copy mutate fitness ranked stage 1
Copy mutate fitness ranked stage 2
Copy mutate fitness ranked stage 3
Copy mutate fitness ranked stage 4
Copy mutate fitness ranked stage 5
Frequency, f(r)
Rank, r
C
Indian cuisine
Sigmoid fit (a = -0.936, α = 0.2535, k = 0.9863)
Random control
Sigmoid fit (a = -1.01, α = 0.1566, k = 0.9907)
Copy-mutate fitness random
Sigmoid fit (a = -1.274, α = 0.1135, k = 1.042)
Copy-mutate fitness ranked
Sigmoid fit (a = -1.186, α = 0.2305, k = 1.006)
P(≤ N_s^R)
N_s^R
D
Indian cuisine
Copy mutate fitness random
Copy mutate fitness ranked
f(N_s)
N_s

The Flavor Graph based on the Indian Cuisine



Low occurrence of pairs

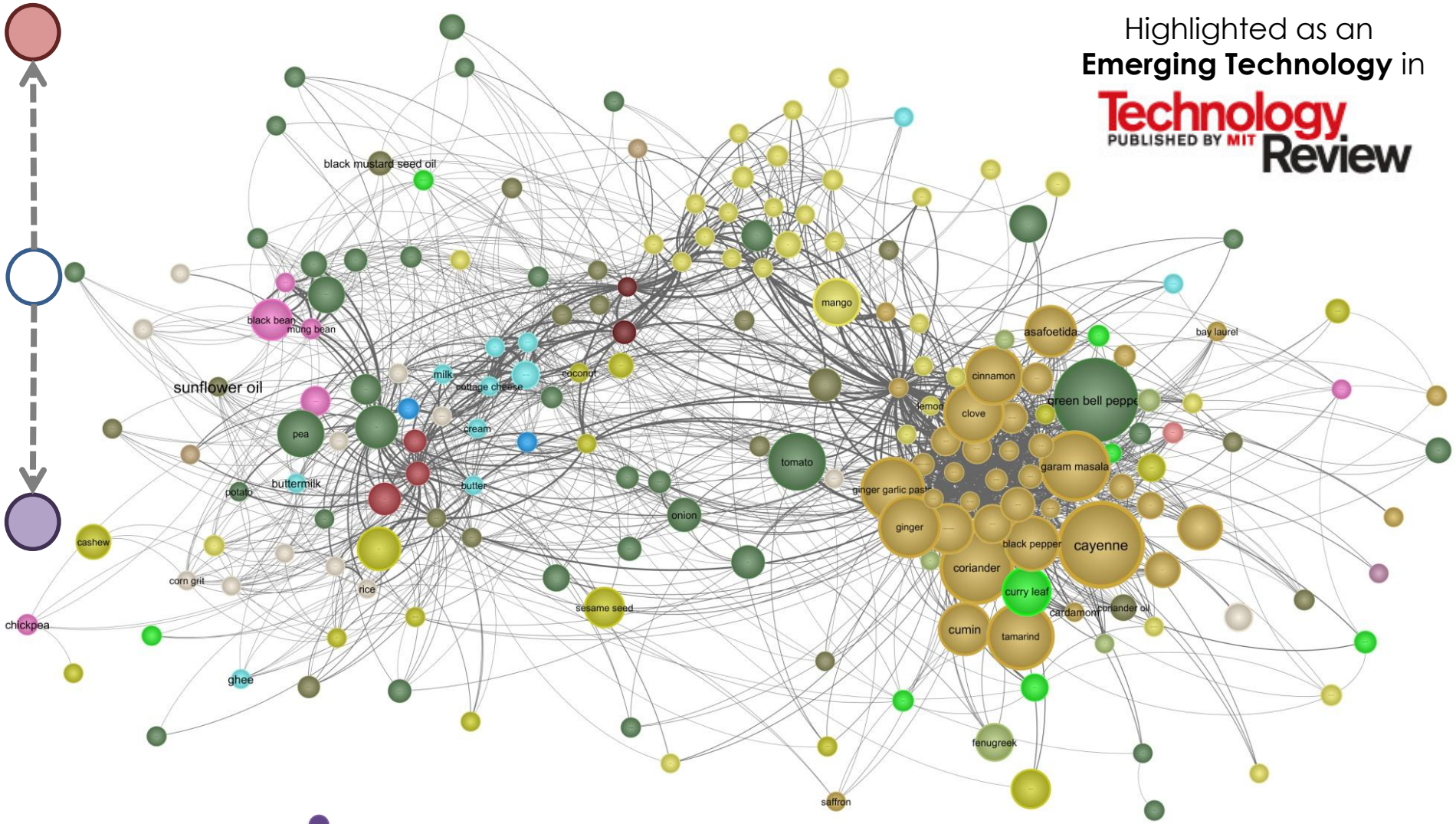
High occurrence of pairs

Low flavour sharing

Medium flavour sharing

High flavour sharing

Food-as-Medicine: Molecular landscape of food

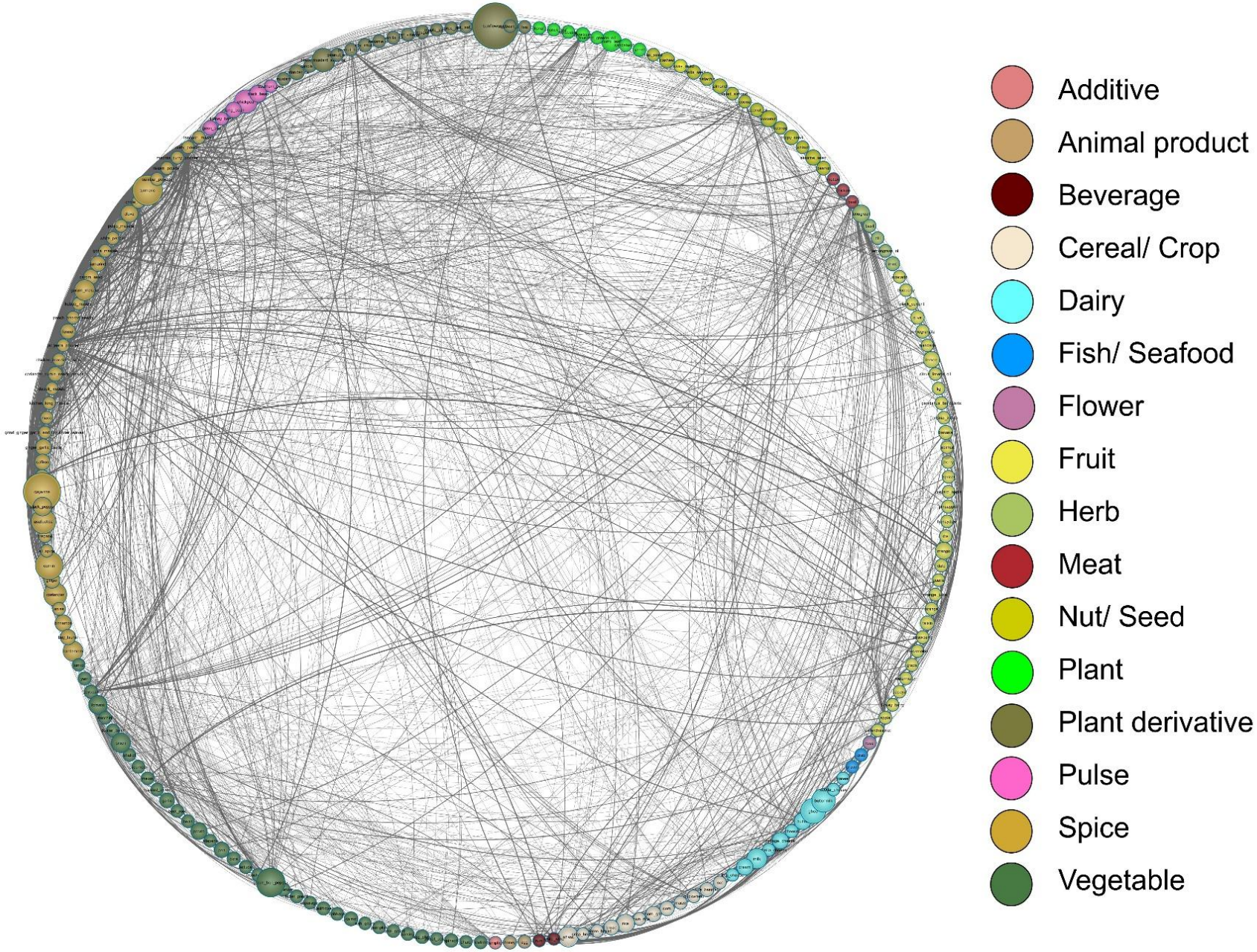


Highlighted as an **Emerging Technology** in Technology Review PUBLISHED BY MIT

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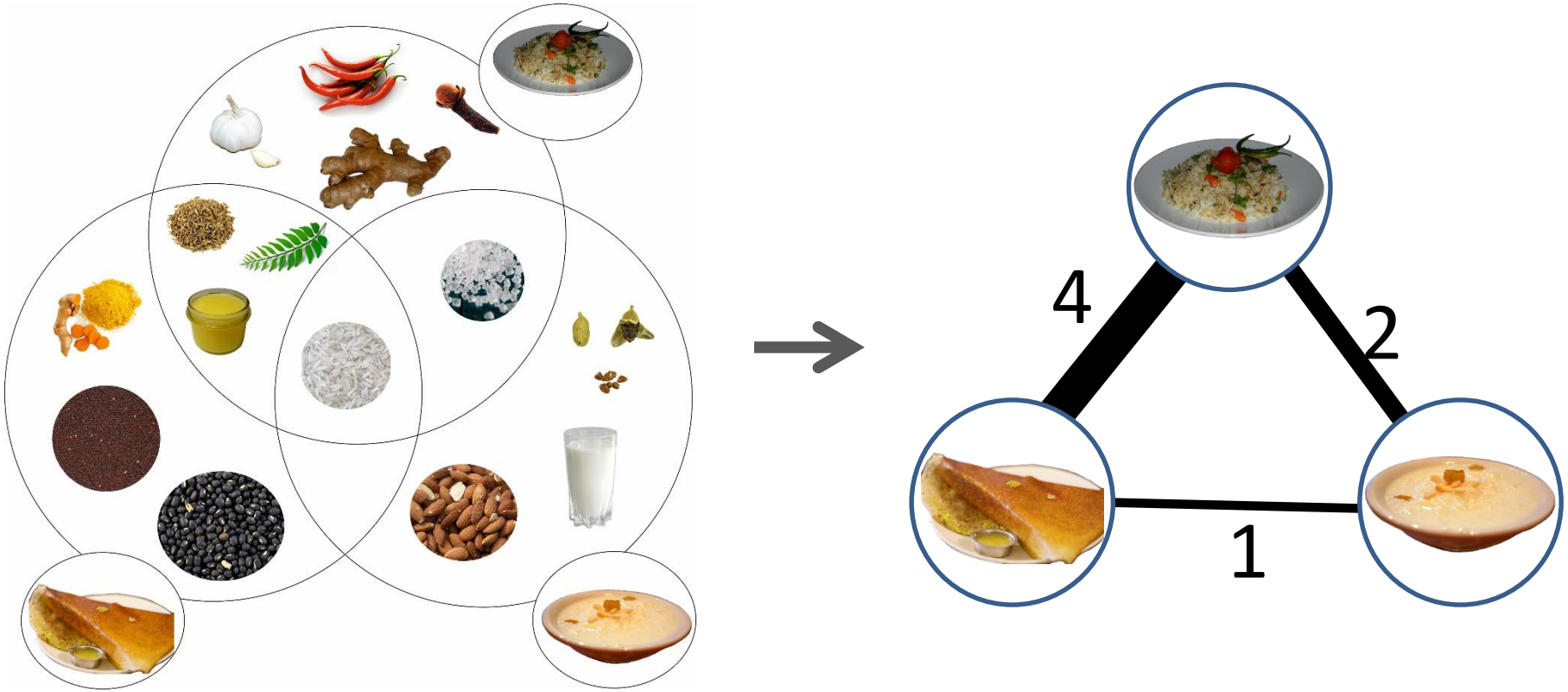
A Jain, NK Rakhi and G Bagler*, PLoS ONE, 10(10), e0139539 (2015).

The Flavor Graph based on the Indian Cuisine

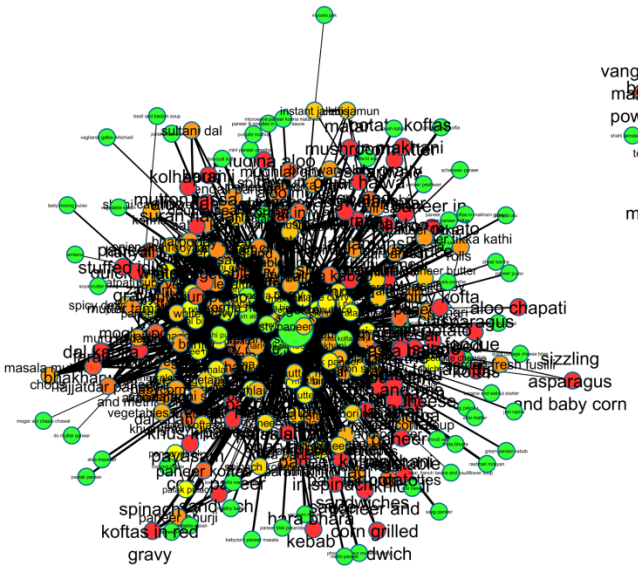


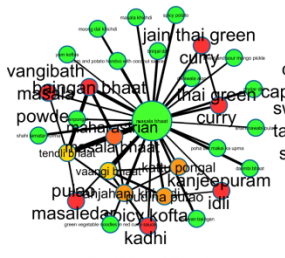
A Jain, NK Rakhi, G Bagler, “Spices form the basis of food pairing in Indian cuisine”, arXiv:1502.03815 (2015).

Construction of Recipe Network

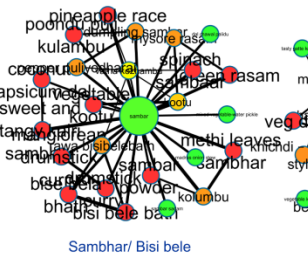


Deciphering Recipe Network of Indian Cuisine

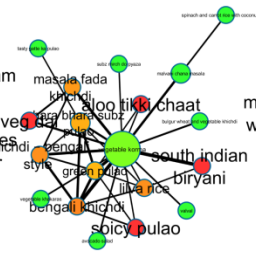




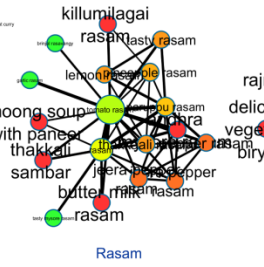
Daal/ Curry/ Bhaat



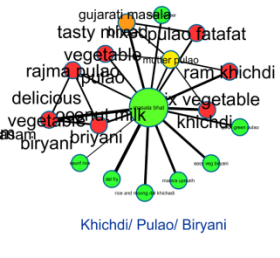
Sambhar/ Bisi bele



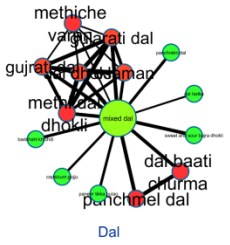
Khichdi/ Pulao/ Spicy veg.



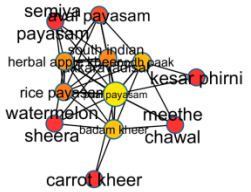
Rasam



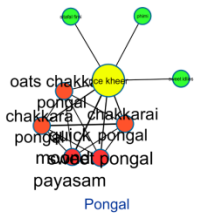
Khichdi/ Pulao/ Biryani



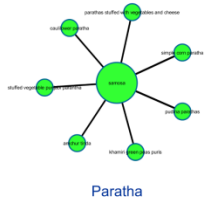
Dal



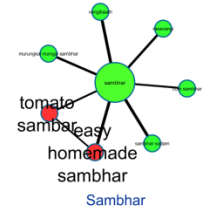
Kheer/ Payasam



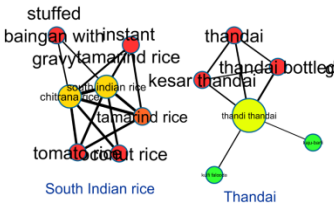
Pongal



Paratha

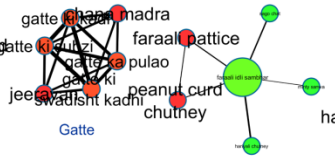


Sambhar



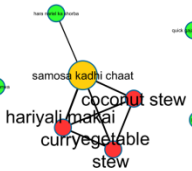
South Indian rice

Thandai

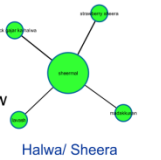


Gatte

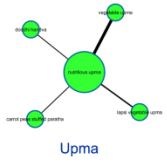
Faraali prep./ Chutney



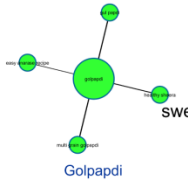
Stew/ Shorba



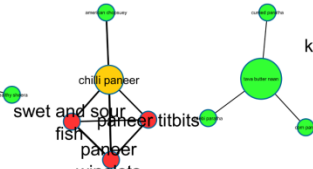
Halwa/ Sheera



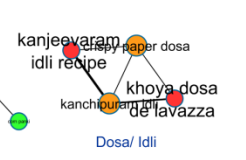
Upma



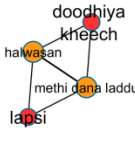
Golpapdi



Fish/ Chilli paneer



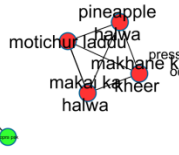
Dosa/ Idli

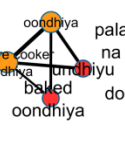












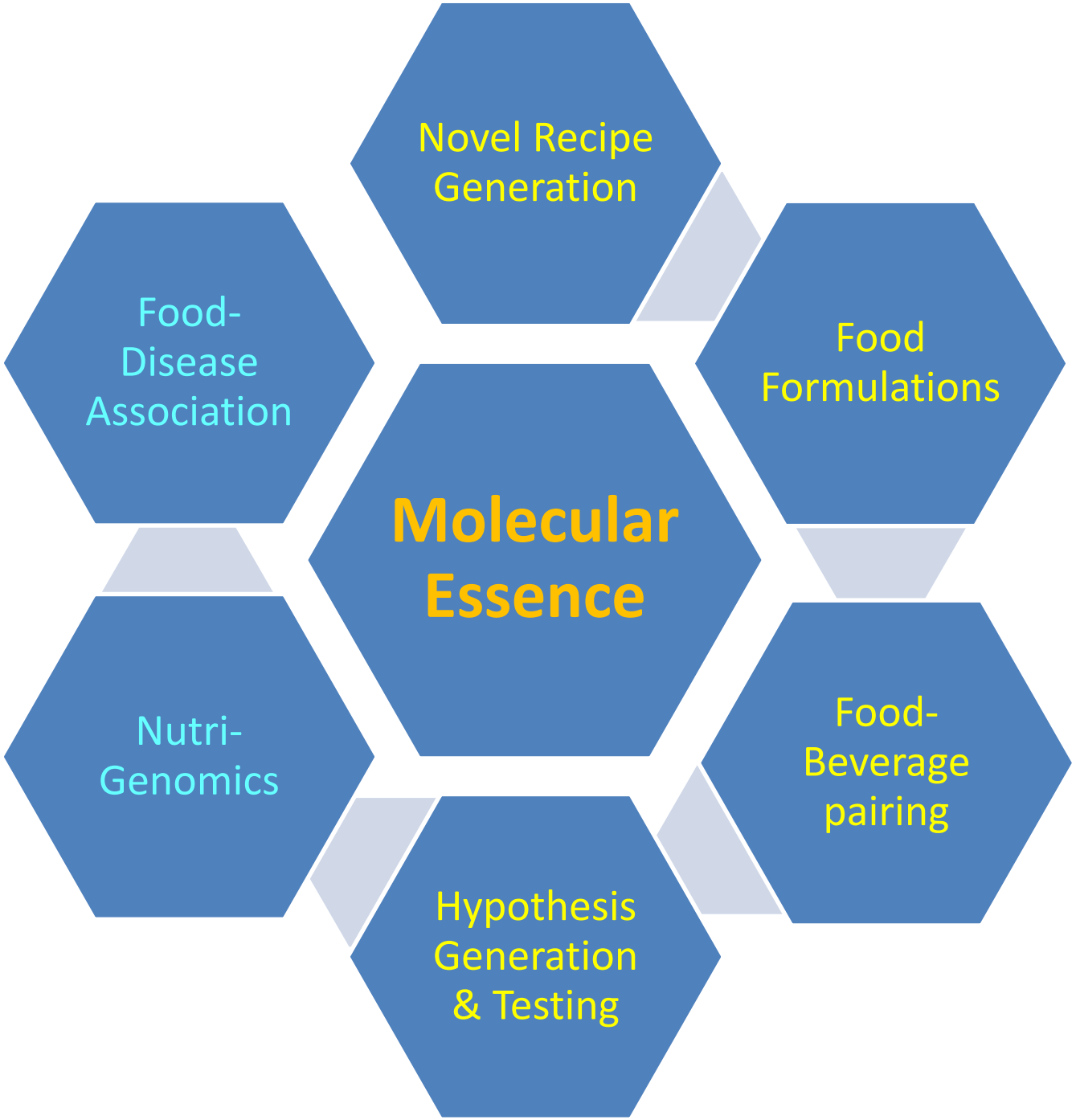









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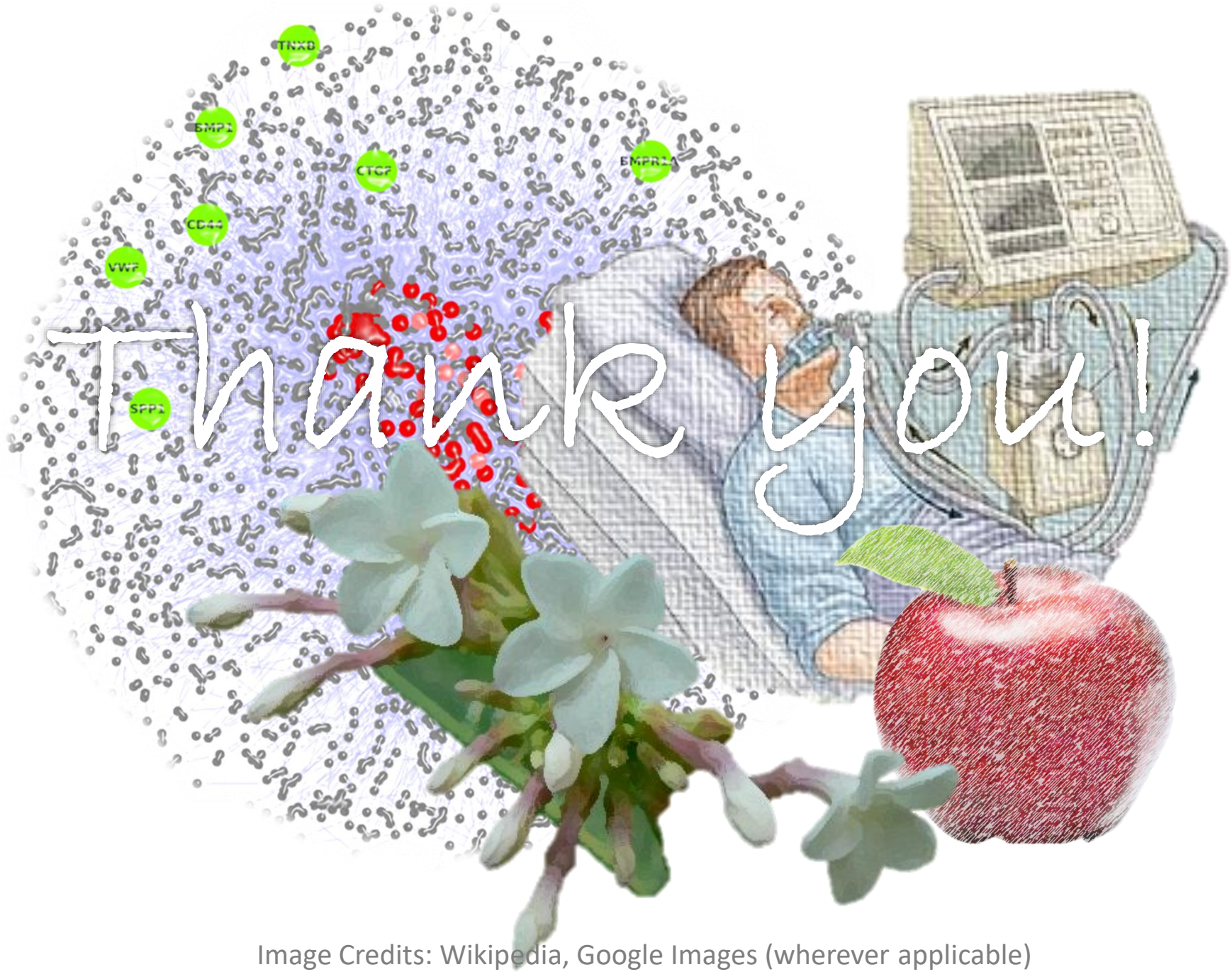


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The molecular essence of Indian cuisine



Ganesh Bagler

Center for Computational Biology, IIIT-Delhi